

1. Welcome to SBS Step Browser

SBS turns a CAD assembly into a narrated, step-by-step illustration: every step is a complete picture of the scene — which parts are visible, where they are, what colour they wear, where the camera stands, what is drawn on top — and SBS animates the journey from one step to the next, speaks the narration, and renders the whole thing to video.

What SBS is for

Assembly and service instructions, training videos, exploded-view walkthroughs, sales demos of mechanical products. You load a model once, then *author steps*: hide a cover, move a screw out, point a camera, write a sentence. SBS handles the in-between motion, the voice, the subtitles, the header, the translation into other languages and the final MP4.

The five ideas you need



Figure 1 — Model and tree on the left, steps (self-contained snapshots) in the middle, transitions and export on the right.

Project

One `.sbsproj` file inside its own folder, plus sidecar folders for audio, render cache, media, languages, exports and backups. The project folder is meant to be moved or copied as a unit.

Model and tree

The CAD geometry (STEP, IGES, OBJ, GLB/gLTF, STL) shown as a tree of folders and parts in the **Tree** tab. You can regroup, rename, archive, add hardware, cables, primitives, notes.

Step

A complete snapshot of the scene: visibility, transforms, colours, camera, overlay, narration. Steps are grouped into *chapters*. A step never depends on the steps around it — you can jump anywhere and the scene is correct.

Transition

What the viewer sees between two steps. SBS diffs the two snapshots and animates the difference according to the step's easing (Smooth / Linear / Instant / Instant fade) and an optional private animation.

Export and cache

Each step is rendered once into a per-step cache. When you edit something, only the steps it touches get a ★ star and are re-rendered. Everything else is stitched from cache.

How to use this manual

The left column is a searchable table of contents. Type anything — a button label, a shortcut, a symptom — and matching topics appear with a snippet. Press to jump to the search box, to clear it. The  **Theme** button switches light and dark.  **Print** produces a PDF with one chapter per page section.

Labels written like **Tree** are the exact words you see in the app. Keys are written like + . Menu paths are written **Edit ▶ Languages....**

2. Getting started

Install and launch

Run `SBS Step Browser Setup <version>.exe`. The installer lets you choose the folder, creates a desktop and Start-menu shortcut, and does not touch your projects on uninstall. SBS is fully offline: the text-to-speech voices, the CAD converters and the video encoder ship inside the installer. Cloud text-to-speech and machine translation are optional and need an internet connection plus an API key.

First launch is slower: the local voice engine warms up in the background (a few hundred megabytes of models are loaded once per session). You can start working immediately; narration playback becomes available when the status bar stops reporting the warm-up.

Two projects at once. Launch SBS again while it is running and a **second window** opens (its title bar says (2)), and a third, up to eight — two projects side by side, as in 3ds Max. Each extra window works from its own copy of the settings: it starts the way the first window is set up, and what you change in it stays in it; the first window is the one that remembers. The licence is shared automatically. **Copy in one window, paste in the other:** overlay items (text boxes, pictures, shapes, tables, video clips) and table cells travel between SBS windows through the ordinary clipboard, look and all. A pasted item that uses a text style, shape style, constant title, pinned position or crop mask is pointed at *this* project's definition: the same brand definition when both projects wear the brand, else one of the same name, else the definition comes along and becomes this project's own (the brand notice will say so). Positions and sizes are in the **export frame**, never the screen: a copy from a 1280×720 project pasted into a 1920×1080 one is fitted by the ratio of the two frames, so it lands at the same place and takes the same share of the picture (text keeps its style's size, as with a brand). Copying from Word or Excel into SBS, and from SBS into them, works as before.

Activating, renewing and moving your licence

First launch. SBS shows **Activate SBS** with this computer's **machine ID**. Press  **Copy** and send your distributor the machine ID together with your email address. They reply with a **password** and a long **key**; paste both and press  **Activate**. No internet connection is needed at any point — the key is checked on your own computer.

Help ▶ Licence... shows who the copy is licensed to, the date it runs until and how many days are left, and the machine ID again with a copy button.

-  **Enter a new key...** — **renew early**. You do not have to wait for the licence to run out: paste the new password and key whenever they arrive. Nothing changes unless the new key is valid, and if it would run out *sooner* than the licence you already have (an older key pasted by mistake) SBS says so and asks before replacing it.
- **Deactivate this computer...** — takes the licence off this computer; SBS restarts and asks for a key. Save your project first: SBS refuses while there are unsaved changes, because an unlicensed copy cannot save.

A key belongs to one computer. It is tied to that computer's machine ID, so deactivating does **not** move the licence somewhere else. To use SBS on a different computer — or after replacing a motherboard or a system disk, which changes the machine ID — install SBS there, copy the machine ID it shows, and ask your distributor for a key for it. Your old key is not destroyed: entered again on the original computer, it still works.

A few days before the licence runs out SBS reminds you at start-up. After that date it shows **License expired** and waits for a new key; your projects are untouched.

Your first project in ten minutes

1. **File** ▶ **New** (**Ctrl** + **N**), then **File** ▶ **Save As...** to give the project a folder. Save early — the render cache, audio and backups live next to the project file.
2. Open the **Files** tab and click **Import**. Pick a STEP, IGES, OBJ, GLB or STL file. STEP and IGES are tessellated on import; large assemblies take a while the first time and are then cached as fast-loading **.sbsobj** files.
3. Look at the **Tree** tab. Rename or regroup folders so that they match how you want to explain the assembly (see chapter 4).
4. Orbit with the left mouse button, pan with the right button or middle button, zoom with the wheel. Press **F** to fit the selection (or everything) in view.
5. In the steps panel at the bottom, add a step. Hide a part (eye icon in the tree, or the Visibility commands in the right-click menu), move another with the gizmo, press **C** to capture the camera into the step. Add the next step and repeat.
6. Type the narration for each step in the text field under the viewport. Press the  button to hear it.
7. Press the **▶** button to play the sequence. Adjust transition easing per step if something should snap instead of glide.
8. Open the **Export** tab, choose size, frame rate and format, and click **Start Export**. Steps are rendered once into the cache; later exports only re-render starred steps.

What a project folder looks like

MyProduct/	
 MyProduct.sbsproj	the project — single source of truth
 MyProduct.he.sbslang.json	one language pack per extra language (see chapter 13)
 audio/<lang>/	synthesised narration clips per language and voice
 render/<lang>/	per-step render cache, frame records, alpha masks
 media/	images and video clips used by overlays
 exports/	finished MP4 / WebM / .sbsproc files
 backups/	automatic backups (Settings ▶ Auto-backup)
 model.sbsobj	fast-load copies of tessellated CAD (offered after the first STEP load)

Figure 2 — A self-contained project folder. Move or copy the whole folder; never move only the **.sbsproj**.

Older projects. Projects saved before the folder layout existed keep their files next to the project (for example a `_rendercache` folder). SBS reads both layouts; new material is written to the new one.

Saving, autosave, backups

`Ctrl` + `S` saves, `Ctrl` + `Shift` + `S` saves as. Saves are atomic — the previous file is only replaced once the new one is completely written. **Settings** ▶ **Auto-backup** keeps timestamped copies in `backups/`; **Back up now** makes one immediately. Closing the window with unsaved changes asks you first.

Rule of thumb. The project file is the truth. Anything you see in the viewport is rebuilt from it when a project loads, so a reload (`Ctrl` + `R` in the app window) is always a safe way to make the live scene match the data.

Taking a project to another computer

File ▶ **Collect Project for Another Computer...** packs everything the project uses into **one .zip** — like Archive in 3ds Max or Collect Files in After Effects: a copy of the project whose paths point *inside* the archive, the CAD models, every video clip the steps use (whole), the brand file, the interface library, and the project's `audio/`, `languages/` and `media/` folders. The render cache is a tick-box, off by default (it is large and the next export rebuilds it). Unzip anywhere and open the `.sbsproj` — nothing to relink.

Where the pictures are. Overlay images, the logo and interface pictures are stored *inside* the project file itself, so they travel with it and never appear as separate files in the archive. Video clips are files and go to `media/`; the voice-over clips go with the audio folder the project names (`audio/`, or the custom folder an older project uses).

The dialog lists every file with where it was found. Each one is **copied** into the archive, **linked** — kept out, the project keeps its absolute path (for a file on a restricted network share that cannot be read here: a computer that can see the share still loads it) — or **skipped**. A missing file can be located with **Browse...** first. `COLLECT-REPORT.txt` inside the archive says what was copied, linked and left out. Files are stored without compression (clips and models do not compress), so the archive is big — that is the point.

3. The interface

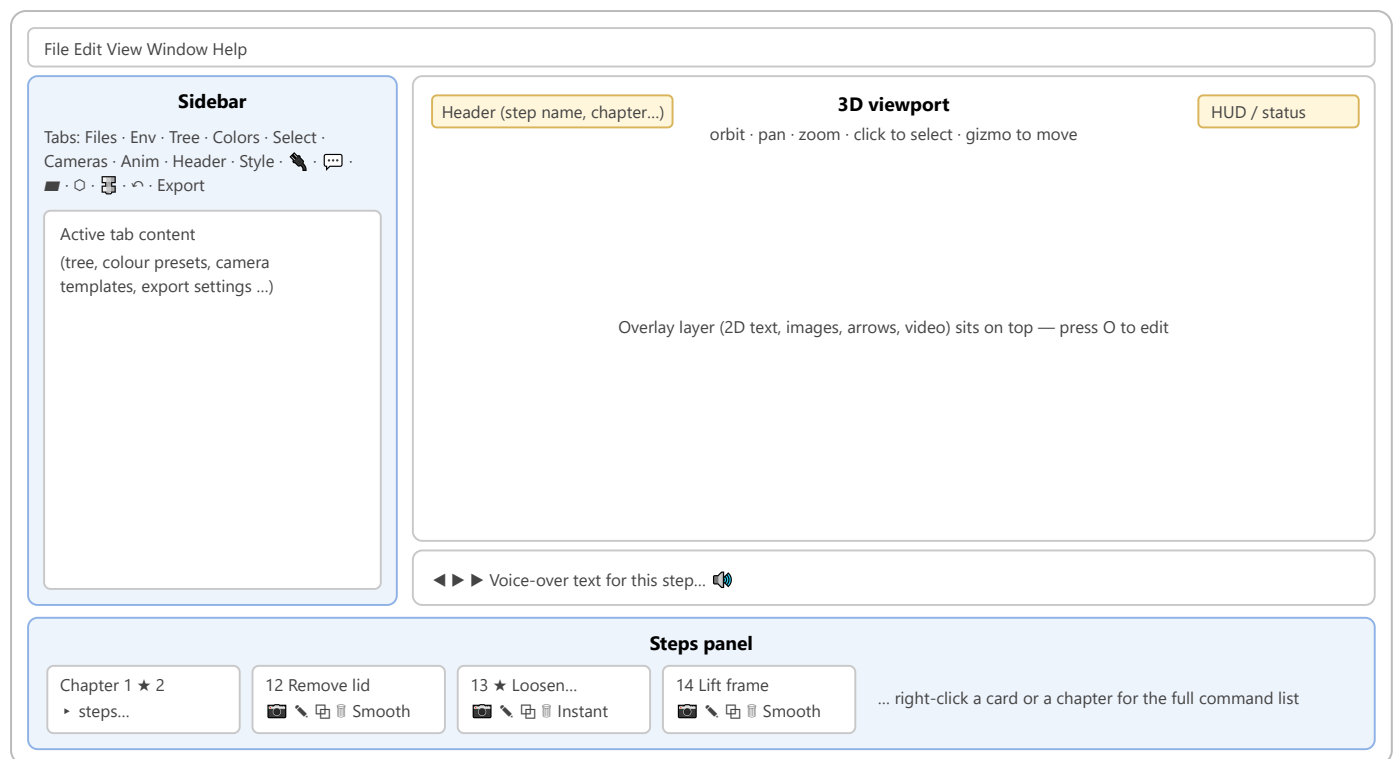


Figure 3 — Layout map: menus on top, sidebar on the left, the 3D viewport with the header and overlay layer, the narration bar, and the steps panel along the bottom.

Menus

Menu	Items
File	New Ctrl + N · Open... Ctrl + O · Save Ctrl + S · Save As... Ctrl + Shift + S · Browse Assets Ctrl + B · Settings Ctrl + , · Exit
Edit	Undo / Redo · Cut / Copy / Paste / Select All · Recover stuck inputs · Apply Camera Template... · Languages... · Model source transform...
Tools	Brand... · Review form... · Document... · Title Manager... · Unify Constant Titles · Clean Up Unused Constant Titles · Rebuild Cascade · Clean up folders... · Scan steps for what is in frame...
View	Zoom In / Zoom Out / Reset Zoom (interface scale) · Toggle Full Screen · Developer Tools Ctrl + Shift + I · Reload Ctrl + R · Hard Reload Ctrl + Shift + R
Window	Minimize · Close
Help	SBS Step Browser Help

Each Edit command is explained in the chapter it belongs to: camera templates in chapter 8, titles and languages in 12 and 13, Rebuild Cascade / Clean up folders / Model source transform in chapter 20, the frame scan in chapter 19.

Sidebar tabs

Tab	What lives there	Chapter
Files	Import models, the list of loaded assets, browse / relink, missing-asset warnings.	4
Env	Background, grid, outline, theme, lights, production-render look (AO, reflections, grade).	18
Tree	The assembly hierarchy: folders, parts, hardware, cables, primitives, shapes, notes. Filters, locks, visibility.	4
Colors	Colour presets and material properties; assign to the selection; highlight colour.	6
Select	Saved selection groups you can load, update or delete.	5
Cameras	Named camera templates; bind steps to them; go to / update / refit.	8
Anim	Animation presets, AL1 / AL2 timing variables, the private-animation editor.	9
Header	The on-screen header: step number / name, chapter, text, image, progress bar, subtitle.	11
Style	Text styles and shape styles for the overlay.	10
	Cables and sockets.	15
	Notes attached to parts, and note templates.	17
	Flat shapes traced from model faces, shape groups, textured planes.	16
	Primitives (boxes, cylinders and so on) placed on faces.	16
	Hardware templates (screws, nuts, washers) and placement.	14
	Undo history — jump back to any earlier state, or clear it.	5
Export	Resolution, frame rate, hold, voice, format, range; the render cache; start the export.	19

Viewport basics

- **Orbit** left-drag, **pan** right- or middle-drag, **zoom** wheel. **Alt** + wheel changes the **perspective** instead of the distance (see 6.4). **F** fits the selection; with nothing selected it fits everything.
- Cameras** ▶  **Refit camera** recomputes orbit distance and pivot after a model rescale.
- **Work camera** (**W**): a scratch camera you can move freely without touching the step's saved camera. Press **W** again to return to the step camera. **C** captures the current view into the active step. While it is on, six **standard views** appear under the button — Top / Bottom, Left / Right, Front / Back, in the axis colours (X red, Y green, Z blue). Hover one to see, in the scene, the plane you would look through and an arrow along the direction you would look; click to switch there orthographically — the view fades out and back in (about 200 ms) rather than flying the camera across, and your

framing is kept. Turning the view (**Alt** + middle-drag) leaves the standard view and brings back the perspective you had before it.

- **Select** by clicking a part. Click again on the same spot to climb one level up the tree (part → folder → model). Double-click a locked folder to temporarily unlock it; it relocks when nothing inside is selected any more. **Shift** adds to the selection, **Ctrl** toggles, **Alt** inverts within the clicked unit. The cursor shows a small badge for the active modifier.
- **Gizmo** appears on the selection: drag the arrows to move, the rings to rotate. **L** switches the gizmo between local and world axes. Type exact values in the numeric fields next to it.
- **Overlay edit** (**O**) switches the mouse to the 2D layer on top of the viewport (see chapter 10). **X** toggles overlay x-ray so you can see the model through the overlay while editing.
- **Global mode** (**Space**): while it is on, a transform you make carries into every step, not only the active one (see 5.5).
- **Esc** cancels a placement tool or picker, closes a popup, or deselects.

Narration bar and playback

Below the viewport: ◀▶ step back and forward, ▶ plays the sequence from the active step, the text field holds the voice-over for this step, 🔊 speaks it. ⬅ / ➡ also move between steps when the focus is not in a text field.

Steps panel

Chapters are the grey headers; steps are the cards. A card shows its number, name, easing, and small buttons: 📷 update this step's camera from the current view, ✎ edit this step's private animation, 📄 duplicate, 🗑 delete. A ★ means the step changed since it was last rendered; a chapter header shows how many of its steps are starred. Right-click anything for the full command list (chapter 7).

Screenshot slot — drop a PNG at the path shown below.

Screenshot: the steps panel with a chapter header, starred cards and the easing dropdown. File:

`docs/manual/img/steps-panel.png`

4. Models and the tree

Importing models

Files ▶ Import accepts `.step / .stp`, `.iges / .igs`, `.obj`, `.glb / .gltf`, `.stl` and SBS's own `.sbsobj`. Several models can live in one project; each becomes a root node in the tree.

- **STEP / IGES** are tessellated by the bundled converter. **Settings ▶ CAD import** sets the quality. After the first successful load SBS offers to save a `.sbsobj` next to the project — the exact same tessellation, loading in seconds instead of minutes. Accept it.
- **GLB / glTF / FBX / OBJ** arrive with their own hierarchy and, sometimes, baked transforms. SBS flattens what it must and records each mesh's resting pose so that steps land on it exactly.
- **Units. Edit ▶ Model source transform...** lets you scale, rotate or offset a whole model at the source level (Metric / Imperial helpers, per-axis reset). Do this once, before authoring steps.

Keep the source files. The project stores the tree and identities, not the geometry. On load, SBS looks for the asset at the recorded path (absolute or relative to the project). If it is missing you get the asset check dialog: **Browse...** to relink, **Override** to load anyway with a placeholder, **Clear All Warnings** once you have fixed the paths. **File ▶ Browse Assets** (`Ctrl + B`) shows the same list any time.

Reading the tree

The **Tree** tab lists the hierarchy. Icons tell the node type: folder, mesh,  hardware, cable, primitive, flat shape, note. Left of every row: the eye (visible in this step), the lock (locked folders select as one unit and refuse edits inside). The filter bar narrows the list: **Sel** only selected objects, **Arch** only archived, **HW** only hardware, **Vis** only visible, **Hid** only hidden, **Fold** only containers (selected parts stay visible). Type in the search field to filter by name. **Unhide all** shows everything in the current step.

Screenshot slot — drop a PNG at the path shown below.

Screenshot: the Tree tab with filters, eyes and locks. File: `docs/manual/img/tree-tab.png`

Organising folders

Drag rows to reparent. Right-click a node for:

Command	What it does
 Rename	Renames the node. Names are what the header, notes and diagnostics show.
 + New Folder Inside	Creates an empty folder under the node.
 Move Selected Here	Reparents the current selection into this folder.
 Collapse	Collapses the branch in the list.
 Clean redundant folders	Removes single-child wrapper folders below this node (CAD exports love those). The project-wide version is Edit ▶ Clean up folders... , which previews the changes, then offers Save first, then remove .
 Delete Empty Folder / Delete Folder (empty first)	Folders with content are protected — empty them before deleting.
 Delete assembly...	Removes a whole loaded model and every reference to it. Asks first.
 Copy tree	Copies the branch as text (for reports or for pasting into a message).
 Archive /  Unarchive	Archived parts drop out of every step and out of the picture, without being deleted. Use it for the fasteners you never want to explain. The Arch filter shows them.
 Isolate /  Un-isolate	Temporarily shows only the node (a viewing aid — it does not change the step).
 Make transformable	Turns a raw imported mesh into a node with its own pivot and transform so it can be moved in steps.

Visibility across steps

The eye icon changes visibility for the *active* step only. To cascade, right-click ▶  **Visibility** (notes have the same row, in the tree and on the balloon):

- ▶  Show on all following steps / ▶  Hide on all following steps
- ◀  Show on all previous steps / ◀  Hide on all previous steps
- ⦿  Show ONLY on this step

Hiding a folder hides everything inside. A part that is hidden by an ancestor shows a dimmed eye. During a transition a part that becomes hidden fades out; one that appears fades in (the fade length is the visibility channel of the animation, chapter 9).

Replace-models and following

Two ways to say "this object should look different in some steps":

-  **Convert to Replace-Model** makes the node a holder for alternative geometry. + **Add to replace...** adds a variant; each step can show one of them.   **Remove from replace-model**

undoes it.

-  **Follow object...** makes object A ride along with object B: wherever B's folder moves in later steps, A comes along, while A keeps its own visibility and colour.  **Stop following...** releases it. Following is the recommended way to attach hardware, labels or tools to a moving part.

Notes on a part

 **Add Note...** attaches a callout to the node;  **Notes on this folder (n)** lists them, with  **Reposition Note...** and  **Delete Note**. See chapter 17.

5. Selecting, moving, pivots

Selection rules

- Single click selects the smallest thing under the cursor (a part). Clicking the same spot again selects its folder, then the model — a climb up the tree. In the tree, click a row or Ctrl / Shift-click for multi-select.
- A **locked folder** (🔒 in the tree) behaves as one object: clicking any member selects the whole folder. Double-click it to unlock temporarily and edit a part inside; it relocks by itself once nothing inside is selected.
- **Colors** ▶ **Select All Meshes** / **Deselect All** do what they say. The Colors tab can also select every part wearing a colour.
- **Select** tab:  **Save...** stores the current selection as a named group;  **Load...** re-selects it;  **Update** overwrites it from the current selection;  **Delete** removes it. Groups survive save/load and are handy for "all the screws of the cover".

Moving with the gizmo

The gizmo appears at the selection's pivot. Drag an axis arrow to move along it, a plane handle to move in a plane, a ring to rotate. L toggles **local** (the part's own axes) and **world**. The numeric fields accept exact values — type a number and press Enter. Every gizmo edit is undoable (Ctrl + Z). If the row's  **Move** (or  **Rotate**) toggle is off, the stored position is not in effect and the part sits at its home position — dragging then starts from where the part actually is, and where you let go becomes the new stored position.

What you move is stored in the *active step* only. Go to the next step and the part is back where that step says. That is by design: a step is a full snapshot. To move a part in many steps at once use Global mode or Group for global edit.

Transform commands (right-click)

Command	Use
 Copy Transforms /  Paste Transforms...	Copy a folder's pose (and every part inside it) in this step; paste onto another folder, or the same one — and pick the steps: just this one, all, previous, following, or the selected steps. The poses are written as they were copied, part by part; the tree in those steps is not touched; the move / rotate / pivot toggles and the pivot's own position and orientation come along exactly as they were on the copy (a pivot that was off stays off). Steps whose tree differs from this one (the folder under another parent, or parts not inside it) are listed first: leave them out, or paste anyway. One undo.
 Reset Move /  Reset Rotation /  Reset All Transforms	Send the part back to its imported pose in this step.
 Global transform...	Apply one numeric move / rotation to the selection across all steps at once.
 Align to surface...	Click a face on another part; the selection's base face is laid tangent onto it.
 Align by 3 points (concentric)...	Pick three points on a circular edge of the target and three on the moving part: the axes are matched (screws into holes, shafts into bearings).
 Fit To	Frames the node in the viewport.

Pivots

Rotation happens around the pivot.  **Copy Pivot** /  **Paste Pivot** move a pivot between parts, and  **Paste Pivot** → **selected steps (n)** pushes it into a step selection.  **Pivot Center via 3 Points...** sets the pivot on the centre of a circle you click three times (hinges, shafts).  **Snap Pivot to Surface...** puts it on a clicked face. When a part rotates around a pivot between two steps, SBS animates the arc, not a straight line — also when the part sits inside a moving parent folder.

Global mode (Space)

Press Space to toggle Global mode (the HUD shows it). While it is on, a move or rotation you make is applied as a *delta* to the selected objects in *every step* the moment you deselect. Use it to fix an assembly-wide placement error without touching each step. Press Space again to go back to per-step editing.

Group for global edit

 **Group for global edit...** wraps scattered parts into a helper folder so that one transform can drive them together across all steps; it shows a dry-run of what it will change before doing it. Pair it with Global mode.

Undo and the history tab

Every edit — tree, transforms, colours, steps, overlay, cables, hardware, settings — goes through one undo stack. `Ctrl + Z` / `Ctrl + Y`, or open the ↶ tab to see the list and jump back several entries at once. 🗑

Clear history frees memory on very long sessions. Loading a project clears the history.

6. Colours and materials

Colour presets

The **Colors** tab holds a list of named presets. Every part wears exactly one preset per step; the preset is what steps store, not raw RGB. Change a preset and every part wearing it changes everywhere. + **Add Color** creates one; double-click to rename; the ★ **Set as Default** preset is the one new parts get.

- **Assign to Selected** applies the highlighted preset to the selection in the active step (a visible colour change between two steps animates as a cross-fade).
- ↺ **Revert to Default** restores each selected mesh to its default colour.
- **Highlight Color** is the colour used for selection highlighting and for parts that a note points at.
- The sort buttons float the colours used by currently visible objects (or by the selection) to the top; the rest turn grey but stay clickable.

Screenshot slot — drop a PNG at the path shown below.

Screenshot: the Colors tab with presets and material sliders. File: `docs/manual/img/colors-tab.png`

Material properties per preset

Property	Effect
Solid color	The base colour.
Opacity	Classic transparency (0–1). Transparent parts are drawn after solid ones.
Solidness	X-ray strength. Below 1 the part becomes a ghost through which other parts show — this is the usual way to reveal an interior without hiding the cover.
Metalness / Roughness	Physically based shading: metals reflect the environment, rough surfaces spread the highlight.
Reflection Intensity · Reflective (SSR contact reflections)	Screen-space reflections for floors and polished plates. Cost is real; use on a few presets.
Strip textures (pure solid color)	Ignore the texture that came with the mesh and use the flat colour.
Outline: Global default / Always on / Always off	Whether parts with this preset get the silhouette outline from the Env tab.
Flat mirror	Turns a planar face into a true mirror (console-driven feature; see chapter 21).

Slider drags are batched into one undo entry per drag.

Colours and steps

A colour assignment is part of the step snapshot. To make a part red only in step 12, assign it in step 12. To make it red from step 12 on, assign it in step 12 and use the step's copy/duplicate to carry it forward, or assign it in each later step. Changing a *preset*'s values (its RGB or solidness) affects every step and stars only the steps where a part wearing that preset is actually in frame (see 19.4).

Deleting a preset is refused while a part still wears it, and the default preset cannot be deleted. Assign the parts elsewhere first.

7. Steps and chapters

What a step stores

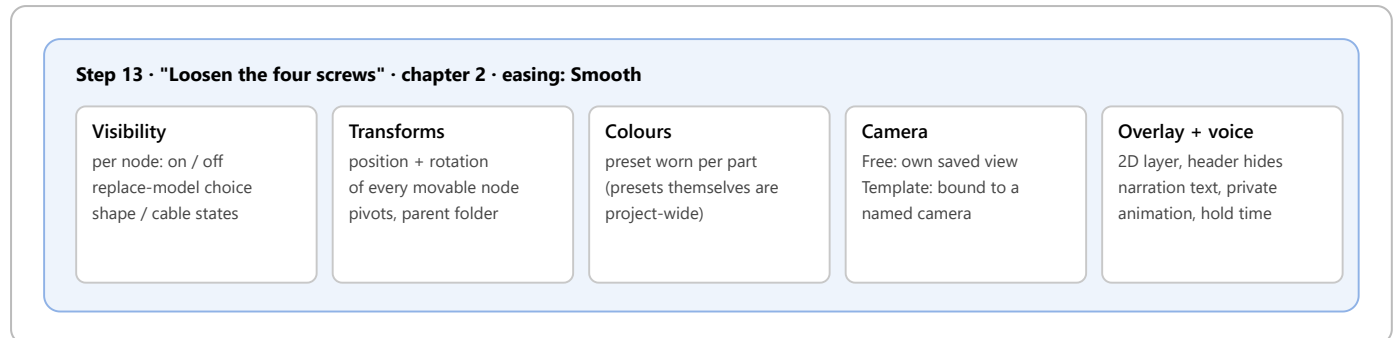


Figure 4 — Everything a step needs to stand alone. Jumping to any step rebuilds exactly this picture.

Creating and arranging steps

- **Add** a step with the panel's add button: the new step starts as a copy of the active one, so you only change what differs. on a card duplicates it in place.
- **Reorder** by dragging cards; drag into another chapter to move it. Multi-select with Ctrl / Shift - click, then drag, copy or delete the whole selection.
- **Chapters** group steps and give the header its chapter number and name. Right-click a chapter header for **Rename...**, **Copy**, **Delete**, selection helpers and **Paste steps into chapter**.
- **Step groups**: **Convert to step group** turns a step into a container whose children play as one continuous clip (one narration, several motions). **Ungroup step** reverses it.
- **Delete** with , Delete or Backspace when the card has focus. The last step cannot be deleted.

Step context menu

Command	What it does
 Rename...	Step name (shown in the header and in exports).
 Auto-name all (by chapter) ·  Auto-name all (from narration)	Name every step "Chapter n – step m" or from the first words of its narration.
 Duplicate ·  Copy ·  Paste under	Copy carries the complete snapshot, including overlay and narration. Paste inserts after the clicked step; with several copied, all of them.
 Import steps from another project...	Opens a second <code>.sbsproj</code> and lets you pick steps to bring in. Works best when both projects load the same CAD (part identities match); missing parts are reported.
 Paste under — from "..."	Steps copied in another SBS window ( Copy on a step or a chapter there). The same import dialog opens with the copied steps already ticked: models this project lacks, colours, cables, rendered clips and the voice-over cache are resolved as for an import from a file. The source need not be saved. A copy made in <i>this</i> window pastes directly, as before.
 Copy overlay ·  Paste overlay — Replace /  Add on top	Move the 2D layer between steps without touching the 3D snapshot.
 Update step camera	Store the current viewport camera in this step (same as ). With several steps selected, all of them.
 Re-render cache segment	Force this step (or the selection) to be rendered again at the next export.
☆ Clear star (treat as rendered) · ☆ Clear all stars	Tell SBS the cached clip is fine even though something changed (19.4).
<cable> · <verb> · <socket> · State n	Cable actions recorded on this step (plug / unplug); ✕ Remove this action deletes one.
☒ Select all steps in chapter · Select all steps & chapters ·  Invert selected	Selection helpers for bulk operations.
 Delete	Removes the step (or the selection). Undoable.

The transition row

Under the active card sits the transition row: **easing**, **camera mode** (Free / Template), **AL1 / AL2** durations and **Smooth reparent jumps**.

Easing	What the viewer sees when arriving at this step
Smooth	Eases out of the previous step and into this one. Default.
Linear	Constant speed. Good for conveyor-like motions and camera slides that continue into the next step.
Instant	The step arrives all at once: a purple INSTANT BLOCK is pinned to the top of the step's animation and holds every channel (camera, objects, colours, visibility, overlay, cables...). Use it for cuts.
Instant fade	Same, but the picture dissolves to nothing and back over AL1 while it cuts.

AL1 and **AL2** are the two animation-length variables in milliseconds (a new project starts with AL1 = 1500, AL2 = 400). Presets reference them instead of fixed numbers: AL1 is the default home of the camera channel, AL2 of the object, visibility and colour channels. Change them on the step to speed up or slow down that one transition; change the defaults in the **Anim** tab.

Smooth reparent jumps: when an object changes folders between two steps but barely moves, hold it still instead of letting it swing through a large arc. The threshold is the distance its visual centre may move before the arc is animated normally.

Instant on the first step of a chapter. Works like any other step. If an old project shows a strange half-transition on such a step, re-pick the easing once from the dropdown — that rewrites the instant block in the current format.

Steps that are videos

A step can also be a rendered clip imported from a render cache (marked "This step IS a rendered video") or contain a video overlay playing alongside the scene. Video steps are exported as-is and cannot be edited in 3D; edit the original project instead.

Imported clip looks soft, blurry, pixelated, low quality — especially its last frame? The clip is the source project's cached segment, and here it is compressed a *second* time when this project exports. At the usual Medium (4 Mbps) that is fine for delivery but thin as a master, and a step whose motion runs straight into the next one (a short or zero hold) ends on a frame the encoder had no time to clean up. Fix, no code: in the **source** project set Export ▶ Quality to **High — 8 Mbps** or **Very High — 16 Mbps**, star the step (or type its number) and **Re-render**, then import it again. The final video here stays at whatever quality you export it at.

8. Cameras

Free cameras

By default a step's camera is **Free**: whatever view you capture with **C** (or  on the card) is saved in that step. Moving the viewport does not change the saved camera until you capture again. Press **W** to enter the work camera — a scratch view for inspecting the model — and **W** again to snap back to the step's saved camera.

Camera templates

The **Cameras** tab holds named views. + **Save current view as template** creates one from the viewport. For each template: ► **Go To** moves the live camera there (bindings untouched),  **Update** sets the template to the current view *and* binds the active step to it,  **Lock** protects it from accidental updates, ★ **Set as Default** makes it the camera new steps get,  **Delete** removes it (bound steps become Free with the last view baked in).

Switch a step's camera mode to **Template** in the transition row and pick a name: from then on the step follows the template, and updating the template updates every step bound to it. **Edit** ► **Apply Camera Template...** binds a whole range of steps at once.  **Refit camera** recomputes distance and pivot to frame the entire scene — useful after a model rescale.

Screenshot slot — drop a PNG at the path shown below.

Screenshot: camera templates with Go To / Update / Lock. File: `docs/manual/img/cameras-tab.png`

Perspective — how wide the lens is

Hold **Alt** over the viewport and turn the **wheel**: the lens opens up or flattens while the camera moves in or out to keep your framing. The subject stays exactly the size it was — only the amount of perspective changes. A badge at the cursor shows a box drawn at the current lens, so you can see the effect before you turn the wheel. **Ctrl** makes the step finer, **Shift** coarser, the same way they change the zoom wheel.

Turn it all the way down and the view becomes **orthographic**: parallel edges stay parallel, like a technical drawing. Turn it up for a dramatic, wide-angle look. A wide lens stretches whatever sits near the edges of the frame — at the default setting a part in the corner is drawn about a third longer than the same part in the middle — so if an illustration looks distorted when the camera is close, this is the control that fixes it.

The perspective belongs to the camera, so it is saved the same way the view is: press **C** (or  **Update step camera**) to store it in the step, or  **Update** to store it in a camera template. Until you do, it is a live

view like any orbit or zoom. Each template's list row shows its lens.

How the camera animates

Between two steps the camera interpolates position, target and zoom over AL1 (or over the camera channel of a private animation). With **Instant** easing it cuts. Orbit pivots and pull-outs stored on a step are honoured — except between two steps that hold the **same view**, where a pull-out would only rock the camera out and back over a shot that never moves, so it is ignored. The first step of such a run still arrives from somewhere else and keeps its own. When two steps hold **different perspectives**, the move is a dolly zoom: the lens and the distance change together so the subject keeps its size for the whole transition instead of swelling and settling at the end. The export renders exactly what the step camera sees at the export aspect ratio; the viewport may show a little more or less on the sides.

Settings ▶ Camera holds the defaults for new projects (field of view, orbit behaviour, default background).

9. Transitions and the animation editor

Channels

A transition is made of channels, each with its own duration. SBS animates whichever channels actually changed between the two steps:

Channel	Animates
camera	Camera position, target, zoom (home: AL1).
obj	Object moves and rotations, pivot arcs, reparent jumps (home: AL2).
visibility	Fade-in / fade-out of parts that appear or disappear (home: AL2).
color	Cross-fade between colour presets (home: AL2).
overlay / overlays	The 2D layer's fade-out of the old overlay and fade-in of the new one.
insert	Hardware insertion (screws driving in, with the reposition pre-step).
cable	Cable morphs and socket plug / unplug.
shape	Flat shapes and primitives appearing, moving, resizing.
notes	Note callouts appearing and following their parts.
narration	Where the voice-over starts relative to the motion.
pause	A time spacer with no channels — a deliberate hold inside a transition.

Animation strings and presets

The **Anim** tab shows the default preset as a string such as `camera(AL1), color(500), visibility(AL2), obj(AL2)`. Channels listed together in one block run at the same time; blocks separated by a phase run one after the other. Numbers are milliseconds; `AL1` / `AL2` resolve at transition time from the step (or the project defaults).

- **+ New Preset** creates a preset; the ★ marks the **Project default** used by steps that have no private animation.
- Presets can be imported from your personal collection (kept in user settings, across projects) and exported to it.
- The three **search zones** filter the preset list by dragging channel chips into them; **Clear all 3 search zones** resets.

Private animation per step

Press  on a card to open the editor for that step only. It shows the phases as blocks; drag channel chips between blocks to reorder what happens first, add an empty time block or a pause block, type a custom duration or pick AL1 / AL2, and remove blocks with  **Remove time block**. The raw string is shown underneath and can be edited by hand.

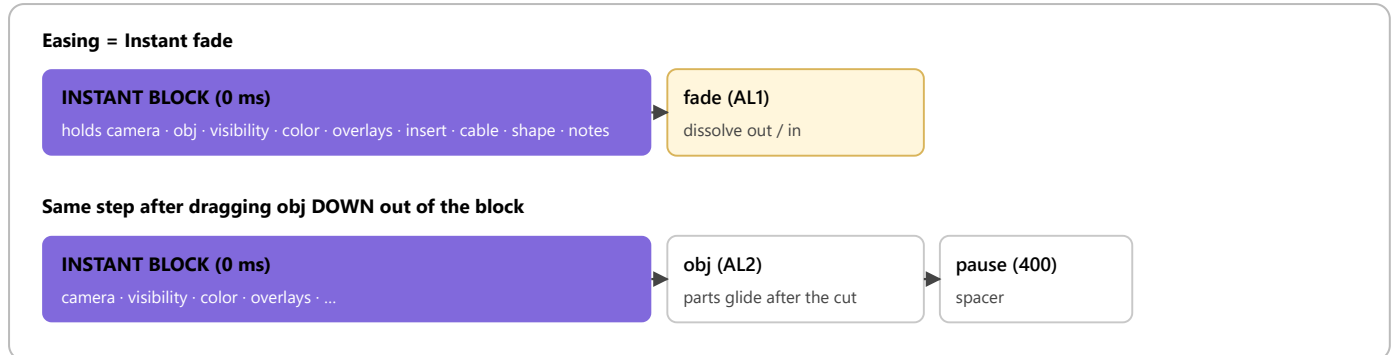


Figure 5 — Instant easing pins a purple block that holds every channel. Drag a chip down out of it and that channel animates after the cut.

Order of precedence. A step's private animation wins over the project default. The easing dropdown edits the private animation (it inserts or removes the instant block). Deleting the private animation returns the step to the default preset.

Holds and narration timing

After the transition the step is *held* while the narration plays. In the exported video the hold is the narration length plus **Export** ▶ **Step hold (ms)**. A pause *inside* a transition is authored with the pause channel, not with the hold.

10. The overlay layer

Entering overlay edit

The overlay is a 2D canvas locked to the picture: text boxes, images, arrows, rectangles, zoom insets, video clips, 3D-anchored arrows. Press (or the overlay button) to edit it; the 3D mouse is suspended until you exit (again, or **Exit overlay** in the prompt that appears when you leave the step). toggles x-ray so the model stays visible under dense overlays while editing (its button sits in the helper row under the overlay bar, next to the magnet). Each step has its own overlay; use **Copy overlay** / **Paste overlay** on the card menu to move it.

Tools

Tool	Notes
Text	Double-click to edit in place: the editor shows the box exactly as it will render — template font and size, bold, effects, fill, rotation. The text toolbar sets font, size, colour, alignment, effects (shadow, outline) and a saved text style from the Style tab. The size box is both a list and a field: click it and the preset sizes drop down; type into it and that is the size (78, 150, 22.5 — pixels of the export frame), Enter applies it to the selected characters; ↑/↓ nudge by one, with Shift by ten. Whatever you last picked becomes the look of the next new text box. Hebrew and mixed Hebrew/English lines read correctly by default (each paragraph follows its first letter); if a line opens with a number or an English term and comes out wrong, right-click ▶ ¶ Text direction ▶ Right-to-left forces it for the whole box.
Table	 Table in the  Assets ▾ menu adds a table to the step. Double-click it and the real table opens over the canvas with its own two-row bar: the top row is the look of the cells you picked (B, <i>I</i>, alignment, size, text colour, cell colour,  to clear), the bottom row is the shape of the table (add / duplicate / delete rows and columns, merge, unmerge, heading row, grid lines, ✓ Done). Click a cell to pick it, drag or hold Shift to pick a block, Enter or F2 to type in one, Tab to move on, Delete to empty what you picked, Esc to finish. Paste a block copied out of a spreadsheet and it fills the cells, growing the table if it has to; Ctrl + C / Ctrl + X copy or cut the picked cells, and pasting them back inside the app brings their look with them. Right-click a cell — on the canvas or inside the editor — for rows, columns, duplicating and merging. A right-click inside a picked block keeps the block, and the menu then offers  Merge the picked cells and  Delete rows 2–4 for exactly what you picked. All eight handles resize the whole table: drag a side or a corner and it is <i>redrawn</i> at that width and height (the text re-wraps, the rows share the height), never stretched; ↑ Let the height follow the rows gives a dragged height back. Inside the editor, each column has a grip above it and each row a grip beside it: click one to pick the whole line, drag it to move that row or column somewhere else (an orange line shows where it will land). The thin handles on the borders between columns and between rows set how wide a column and how tall a row is; ↔ Even and ↑ Auto put those back. Pictures go in cells: right-click a cell ▶  Picture in this cell..., or drag an image file onto a cell, or paste one from the clipboard. The row is given a height so the picture has somewhere to sit — drag the row's border to make it bigger. The picture is stored shrunk to a generous ceiling, so enlarging the cell, the column or the whole table later still draws it sharply, and it costs a fraction of the original file. Every change is its own undo step, and while the editor is open Ctrl + Z walks back through the table only. Only the table's text is stored in the step, so it costs no more than a text box.
Image	PNG / JPG / SVG from disk; copied into <code>media/</code> . Right-click ▶  Replace picture... swaps the file behind the item and keeps everything else — where it sits, how big it is, how it is turned, its crop mask, its pins, its layer order (a zoom crop is carried across in proportion, so the same part stays framed even at a different resolution). Pictures and clips squeeze: all eight handles resize freely, so you can deliberately distort one — hold Shift to keep the proportions instead. ↔ Restore proportion puts the shape back without moving the item or changing how much room it takes (the area is what is preserved), while ↺ Reset to natural size goes all the way back to the file's own pixels and centres it. Also  Add zoom (an inset crop),  Zoom level 1× ... Custom.  Square up (perspective)... straightens a phone photo of a screen or panel: drag the four

	corner dots onto what should be a rectangle, pick its proportion (Auto, 16:9, 4:3...), press Enter — the whole picture is re-formed so that quad is flat and square, and a crop mask shows just that rectangle. When the thing you can trust (a screen, a label) is smaller than what you want to keep, pull the orange square handles on the four edges: each moves its edge out (or in) under the same perspective, and that frame — not the four dots — is what the mask shows (the rest of the photo is still there behind the mask: scale the picture down under it for context, or edit the mask). One undo. ↶ Back to the original file reads the untouched photo again from disk (the original pixels are not kept in the project). 📁 Save to interface library... writes the picture as it is now into the interface library folder, so Add interface... offers it as a reusable panel.
Shapes and arrows	Rectangle, line / arrow (arrowhead on tail / head), corner radius, fill and stroke from the shape toolbar or a shape style . ➕ Reposition HEAD... / TAIL... re-anchor an arrow by clicking.
 3D arrow	An arrow whose head is pinned to a point on the model: it follows the part across steps and camera moves.
Video	A clip from disk (never copied into the project file). Right-click ▶  Replace clip... puts a different file behind the same frame, keeping its place, size, masks and pins (the trim starts as the new clip's full length);  Trim & audio... for in/out points, mute and the clip's Fade in — how long its first frame fades in before it starts moving (the pause before the clip): AL1 / AL2 follow the Animation settings, or type milliseconds (or 1.5s); the default is 400 ms. It applies when the step's animation has an overlay block, which every preset has. The pause after the clip is simply the next step's transition, which fades the frozen last frame out. The clip plays in the viewport and is included in the export.
Image sequence	A flipbook of frames with timing (the sequence editor: + Add frame... , seconds per frame, Clear sequence).

Common commands: 📄 **Copy**, 📄 **Paste**, 📄 **Paste in place**, 📄 **Duplicate**, 🗑️ **Delete**, 📄 **Arrange** (Bring to front / forward, Send backward / to back). Arrow keys nudge; Shift while rotating levels to 45° / 90°; the rotation field accepts a typed angle.

Selecting with a box. Press on an *empty* part of the picture and drag: a box follows the pointer, and on release what it caught becomes the selection. The keys are the 3D view's box-select, to the letter — a small badge beside the pointer shows them while you drag:

Hold	The box...
	catches whatever it touches (📄) and replaces the selection
Ctrl	catches only what lies fully inside it (📄)
Shift	adds what it caught to the selection (green +)
Alt	removes what it caught from the selection (red -); Alt wins over Shift

Ctrl combines with Shift or Alt. "Touches" means the item's true outline, not its bounding box: a box drawn in the empty corner beside a diagonal arrow or a rotated rectangle does not catch it, and a masked image counts by the window you can see. Esc while dragging drops the box and leaves the

selection as it was. A press and release without a drag is still a click on empty space (it deselects; with **Shift** or **Alt** held it leaves the selection alone). The box selects overlay items only — header items, which live on every step, are added on purpose with **Shift**-click.

Moving several items together. **Shift** - or **Ctrl** -click adds items to the selection — overlay items and header items can be selected together — and dragging any one of them moves the whole selection as **one rigid group**: the items keep their places relative to each other however fast the drag starts, and the move is **one** undo step. What stays put inside a selection: pinned items and constant text boxes (a status line says how many) and 3D-anchored arrows, whose place belongs to the model. An interface panel in the selection brings its bonded shapes along. The group follows the verdict of **the item you hold**: drop it over an interface panel and every carried shape bonds to that panel, drop it over none and they are set free — except shapes whose own panel travelled with them, which simply stay bonded. A carried linked shape updates its link like a dragged one. Holding **Shift** or **Ctrl** while pressing on a selected item takes it *out* of the selection first, so it then drags alone. While a pin is armed for **Set as new position**, the drag moves the pin only.

 **The magnet.** While you drag an item, its left edge, centre and right edge — and its top, middle and bottom — **stick** to the same lines of the other items, of the header items, and of the picture itself (its four edges and its centre). A thin guide shows what it is standing on — **magenta** for a line of another item, **cyan** for a line of the picture itself (an edge or its centre); two guides at once mean it sits on a vertical and a horizontal line. While you drag or resize, the floating panel that serves the item fades to a trace so it hides nothing. **The part you hold is the part that sticks**: hold an item by its left edge and the left edge snaps, hold it by the middle and its centre does; only when that line finds nothing nearby do the others get their turn. A selection of several items moves — and snaps — as **one box**, by its outer edges and its centre, so the group keeps its shape. Hold **Alt** while dragging to let go of the magnet for that moment (fine placement right next to a line). The **helper row** under the overlay bar holds  **X-ray**,  **Magnet** (**M** switches it on and off) and a  that opens the helpers' settings: what the magnet sticks to (other items / the picture) and how far it reaches (in screen pixels, so the pull feels the same at any zoom). These are *your* settings on this machine, not the project's; nothing of the magnet is rendered or saved. 3D-anchored arrows are not snapped to (they follow the camera), and a masked picture counts by the window you can see.

 **Lines and arrows: points, heads and bends.** A line and an arrow are the same thing — a line is an arrow that wears no head. **Click** one and it is a unit: move it, turn it, scale it. **Double-click** it and you are editing its **points**: a dot appears on the head, on the tail and on every point between.

- **Drag a dot** to move that point. The magnet works for points too: a point sticks to the other items, to the picture, and to the *other points of its own line* (so a bend lines up with an end) — never to another line's points. **Shift** keeps the point on one axis, **Alt** lets go of the magnet.
- **Double-click the line** where there is no dot: a new point, right there.
- **Double-click a middle dot**: that point is deleted.
- **Double-click an end dot**: the arrowhead at *that* end goes on or off (a filled end dot wears a head). Both ends are independent — two heads, one, or none; give a line's end a head and it has become an arrow.

- Drag the line itself and the whole thing moves, dots and all. **Esc**, a click on empty space or on another item puts the dots away.

Right-click a line ▶ **~ Bend** chooses how it goes through its points — the cables' three ways: **Corners**, **Smooth (Bézier)** (a curve through every point) or **Fillet (rounded corners)**, with a tight / medium / wide fillet — the cables' rule: the rounding starts the same distance before every corner, so a gentle turn gets a wide arc and a sharp one a tight arc (a corner that has no room takes less). New lines are born with Fillet, so the first point you add bends the cable way. Every point move, added or deleted point, head and bend is one undo step; linked lines carry their points, heads and bend to every copy. 3D-anchored arrows keep their own rules (their ends belong to the model).

⬠ **Shapes into polygons. Double-click** a rectangle, a triangle, a circle or an ellipse (or right-click it ▶ ⬠ **Convert to polygon (editable points)**): it becomes a polygon of free points and its dots appear at once — same place, turn, squish, colours, thickness, style, bond and pin, same place in the stacking order. From then on it is edited like a line: double-click it for its points (drag one, double-click an edge for a new one, double-click a point to delete it — a polygon keeps at least three), right-click ▶ **~ Bend** for corners, smooth or fillet. A rectangle's rounded corners become a fillet of the same size; a circle or an ellipse becomes eight points with Smooth, so it still looks round until you move one. The conversion is one undo step. A linked shape leaves its link (the other copies keep theirs).

⬆ **Moving straight.** While you are dragging — the mouse button still down — hold **Shift**: the move stays on **one axis**, straight along X or straight along Y, measured from where the item (or the group) stood when you pressed. Whichever way you have gone further decides, so you can swing from one axis to the other; let go of **Shift** and the item is free again. It answers at once, even with the mouse standing still, and it works with the magnet on or off (with it on, the magnet then pulls along the free axis only).

Squishing shapes. Shapes squish the way they do in a drawing program: the *shape* changes, its **outline stays as thick all round** as its Thickness says. Squash a circle and it becomes an ellipse, squash a triangle and it stays squashed (it used to spring back to an equal-sided one), stretch a line or an arrow and its **points** move — the bends (fillet, smooth) are worked out again from them, so rounded corners stay round and the line stays even, already while you drag. The frame around several items always stands upright, even when the items in it are turned, so stretching it on one side squashes turned items *along the frame* — a turned rectangle becomes a slanted one (a parallelogram), a turned circle an ellipse turned its own way. Hold **Shift** (or use a corner on images) to resize in proportion instead — and hold **Shift** while resizing a circle or an ellipse to make it a **perfect circle** again, however squished it was. One resize or one rotation of a group is **one** undo step, and undo brings back everything.

Links, constants and pins



Make linked shape... / Link to

Several items across steps share one definition: edit one and every linked copy follows. **✂ Break link (make unique)** detaches a copy; **↺ Reset this one to the link** re-syncs it; **🏷 Rename link....**



Make constant text box... / Attach to constant

A text box that appears in the same place on many steps with the same style — chapter titles, part callouts. The anchor (⌞ top-left / ⌞ top-right) decides which corner stays put when the text length

changes.  **Snap back to constant position**,  **Set as new position (all steps)**,  **Detach from constant (this box only)**. Constant *titles* (their texts and translations) are managed in **Edit ▶ Title Manager...** (12.5).

Make pinned position... / Pin to position

The same idea for images, shapes and video: a named position in the frame. A selected pinned item (and a constant text box) wears a red pin at its anchor corner — on the crop window when the image is masked. It cannot be dragged or nudged: the definition owns its position. Trying opens a small bubble on the item.  **Reposition (all steps)** in that bubble (or right-click ▶  **Set as new position (all steps)...**) lets you drag or nudge it freely; the bubble follows and offers  **Set new position** (the definition is rewritten, every step follows, undoable) or  **Reset to original**. Leaving the step before choosing cancels the move and says so in the status bar. To move only one copy, right-click ▶  **Unpin (this item only)**.  **Snap back to pinned position** returns a stray copy to the definition.

Crop masks

A mask crops an image or video to a rectangle fixed on the canvas (a window through which the picture is seen). Right-click an image ▶  **Add mask...** ▶  **New rectangle mask...** (private to this image) or  **Use a shared mask...** from the project library.  **Edit mask** (moves every image using a shared one),  **Remove mask (show the whole image)**,  **Use another mask...**,  **Set as global mask...** publishes a private mask to the library. Going the other way, an image following a shared mask offers  **Make this mask mine** — it keeps the exact shape, place and angle and simply stops following the shared one — or  **Make it mine and edit it...**, which opens the editor on a copy so the shared mask is left untouched.  **Manage masks...** opens the **Masks & pinned positions** panel, which lists every mask and pin with the steps that use them and jumps to each.

Turning a mask. In the mask editor the rectangle has a  knob **above it and to its left** — two, because the editor's own Apply/Cancel bar sits over the top one. Either turns the rectangle about its centre; hold  **Shift** to land on 15°.

The mask follows its picture. Move, turn or grow a picture and its own mask comes with it, so the framing you cut survives the edit instead of having to be cut again. Hold  **Ctrl** to invert that for one gesture: the window stays exactly where it is while the picture moves under it.

It follows position, rotation and **even** scaling — not **squishing**. Making a picture taller or wider is a deliberate distortion of the picture, so the window is left alone; growing it evenly is "bigger, same thing", and the window grows too.  **Ctrl** is its own modifier layered over  **Shift**:  **Shift** keeps doing its job (snap the angle, keep the proportions) and  **Ctrl** independently decides whether the mask comes along, so  **Ctrl** +  **Shift** gives you both.

A note stays in the status bar for as long as the gesture lasts and changes as you press and release  **Ctrl**, so it always says what is true right now. Everything is measured from where the gesture *started*, never accumulated, so  **Ctrl** can be pressed or released halfway through and the mask lands exactly where it would have either way. One undo step covers the edit and the mask together. Only a **private** mask follows — a shared one belongs to every image using it.

Hands (🖐️ tab)

A hand with a grip — drawn as a bone rig (palm, five fingers of three bones, a forearm), so a real hand model can be skinned over it later. + **Add hand** asks left or right and puts the hand in the tree like any object (drag it into a folder, hide it per step, move it with the gizmo). Then pick **what it holds**: a handle or bar, a pistol grip with its trigger, a pinch, a flat push, a knob — or relaxed. Each grip shows a translucent **ghost** of the thing held, with three numbered points on it (the tab names them: "one end of the handle", "the trigger"...).

 **Align (3 points)** is how the hand lands: click, on the real part, the three points the ghost names, in order (the snap picker grabs vertices, edges and faces; Backspace takes a point back, Esc cancels). After the third click the hand moves so its ghost sits exactly on the part, and the ghost hides (the **ghost** box shows it again). **Closed** blends the relaxed hand into the full grip. **Release at this step** lets go: the fingers open (**Open**), and the hand is a unit under its own transform. Everything is stored per step. **Length** is the hand's size in mm, wrist to middle fingertip.

Fine-tune — double-click the hand (or **Fine-tune fingers...**): a handle appears at each fingertip and a yellow one down the forearm. Drag a fingertip handle to pin that finger where you drop it (it stays on the part it lands on and bends to reach it; on the drawing, cyan = pinned, click to unpin), or click a finger on the drawing and then the spot on the part. Drag the forearm handle to swing the arm. **Reset** unpins everything.

Adjust the grip — when a pose's prop does not sit right on the real part,  **Adjust grip** holds the prop still while you move and rotate the hand with the gizmo and drag the fingertips; ✓ **Done** (or Esc) sets the new seating, which **Align (3 points)** then keeps. **Reset offset** puts the prop back where the pose lays it.

Saved grips — ★ **Save grip...** keeps the fingers as they are (pins baked into angles) together with the prop's seating, under a name, and asks which prop aligns it; it appears among the poses on this computer. A hand that uses a saved grip carries its own copy, so the project opens the same anywhere; **Delete grip** only removes it from the library. When the hand already uses a saved grip, **Save grip...** offers to update it or save another; a right-click on a grip's tile updates it from the hand, renames or deletes it. The library belongs to this computer, not to the project. **Copy / Paste grip** — right-click a hand: everything from the wrist up (pose, fingers, pins, the prop and its seating, release, the forearm point) travels to another hand or to the same hand at another step; the hand itself does not move.

Skin (a real hand) — a skinned hand ships with the app and is on by default, with three textures to pick from in the Skin card (**Skin**, **Blue glove**, **White glove**); **Procedural** shows the bare capsule rig, **Built-in** brings the shipped hand back. Underneath, the hand is a bone rig on purpose. **Export rig...** saves it as an `.fbx` (or `.glb`): right hand, at rest, with a proxy mesh already skinned to the bones. Open it in your modelling app (3ds Max: in the FBX import settings turn *bones as dummies* off, or run

```
FBXImporterSetParam "ImportBoneAsDummy" false
```

in the MAXScript listener first, so the joints come in as real bones), skin a modelled hand to those bones — keep the bone names, do not re-parent them — and export an `.fbx` (binary or ASCII, FBX 2011 or newer) or a `.glb` (no Draco compression). A diffuse texture is picked up when the file has one: export with *Embed media*, or keep the image next to the file.

Load skin... puts it on every hand, left hands mirrored, sized to each hand's length; the hand's material and every SBS look (X-ray, fades, outlines) apply to it. The skin is a setting of this computer, not of the project. ✕ returns to the procedural hand.

First version. The ghost props are simple shapes; the poses are approximate. Both hands at once, a real skinned hand and the render-cache key for hands come later.

Text and shape styles (Style tab)

+ **New style** (text) and + **New shape style** capture font, size, colour, fill, outline and thickness under a name. Apply from the toolbars' style dropdown. **The eyedropper**  **on a text style row** copies the look of a text box into that style: click it, then click any text box on the canvas (edit mode or not) — font, size, colour, bold/italic/underline, drop shadow and outline, and the box's backdrop come across; a box with mixed formatting lends its first letter's. Every box bound to the style follows at once; Esc cancels the pick.

Colours, anywhere on screen. A click on any colour swatch opens SBS's colour dialog: the colour square, the hue strip, HEX / RGB / HSL fields, and an **eyedropper**  (the pipette). The eyedropper takes a snapshot of **every display** — SBS's own window included, so its interface is pickable too — shows the snapshots edge to edge, and the loupe under the pointer reads the pixel: on either screen, in a browser, a picture viewer, anything that was on screen. Click to take it, Esc cancels. A **right-click** (or **Alt** + click) on a swatch goes straight to the screen pick. **Photographing the screens takes a moment** (about half a second on two displays): meanwhile the pointer shows the busy ring, the eyedropper button spins and clicks inside SBS are held — wait for the picker to appear rather than clicking again.

Save Setup / Load Setup write and read the whole style set as a file, so a house style travels between projects. Changing a style re-styles every item using it and stars the steps where those items are.

Deleting a style that is in use asks which style takes its place: every text box on every step, header item and constant title bound to it moves onto the style you pick (one undo). **No style — unbind them** is on the same list, as an explicit choice: the boxes keep the formatting they have, header items fall back to the default look. Shape styles work the same way. A style nothing uses just asks for a confirmation.

Interface panels

An *interface* is a reusable image panel (a control panel, a display) whose buttons are traced as bonded flat shapes and whose crops act as zoom windows. Interfaces are authored from the overlay and the  tab; their library folder and default pose are stored in the project. Ask for the interface guide if you need this feature — it has its own workflow.

11. Headers

The header is the strip of live text drawn over the picture on every step: step number and name, chapter number and name, free text, an image (logo), a progress bar, and the subtitle line. It is one shared layout, evaluated per step. The progress bar is not text, so the default style's colour does not reach it: its own **Fill colour**, **Track colour** and **Track opacity** sit in its editor, beside its position and size.

Header tab

Add items with + **Step #**, + **Step Name**, + **Chapter #**, + **Chapter Name**, + **Project Name** (the project file's name, so name the file after the procedure), + **Text**, + **Image**,  + **Logo**, + **Progress Bar**,  **Subtitle**. Each item has X / Y (px), Width / Height, Font size, Color, Direction (L / C / R alignment), Language (which language's text it shows) and, for images, **Replace image....** Reorder with ↑ ↓, hide with the eye, delete with X. **Save Setup / Load Setup** store the layout as a file.

The project's logo.  + **Logo** adds a header image like any other — placed, sized, replaced the same way — but it is *the* logo: there is one per project, the printed document's header shows it, and a brand carries it as the logo. Pressing  + **Logo** again on a project that has one **replaces** the logo in place — it asks first — keeping its position and width and taking the new picture's proportion, so the logo never moves and never stretches. Any header image can become the logo with **This is the project's logo** in its editor; ticking it on another image moves the flag. Hiding the logo with the eye keeps it the logo: whether it is shown over the film and whether the manual carries it are two separate decisions. On any step,  **Assets** ▾ ▸  **Project logo** in the overlay toolbar puts the logo on that step as an image, at the place and size it has over the film — a copy, which replacing the logo later does not change. A project with no logo yet can pick one right there; it becomes the header logo and lands on the step. A project that never defined a logo keeps the old rule — the document uses the first visible header image, and that image's row in the Header tab says so.

The **Text** item's placeholder reads "Header text (used in Default / template modes)": it is the fallback when a step has no own text; it can be translated with the  button once language packs exist.

Spotlight at this step

"Before I insert the screw I want it up close to the camera." Right-click a folder, a screw, a primitive or a shape ▸  **Spotlight at this step**. On **this step only** the object sits *in the picture* rather than in the world: by default in the middle of the left third of the frame, a third of the frame high, showing the face it showed when you switched it on. Orbit the camera on that step and it comes along; the next step puts it back where it lives — or, for a screw, the insertion animation takes it from there into its hole.

Move or turn it with the gizmo and that place is kept. The **white block** beside the gizmo is the **dolly**: drag it **up** to send the object farther away (smaller on screen), **down** to bring it nearer (bigger) — always along the line from the camera through the object's centre, so it never shifts sideways. It looks like a scale; it is a distance.  **Reset the spotlight place** puts it back on the default;  **Stop the spotlight at this step** switches it off and the object is **back where this step had it** before: the spotlight is a layer over the step's own pose — absolute, but not destructive. **Right-click the white block** for the spotlight's own menu:

reset, re-read the place from where the object is now, **make this place the project default** (every spotlight switched on from then starts there — saved with the project), stop.

In the custom animation the move has a capsule of its own:  **spotlight**. Drag it into the sequence and the spotlighted objects (and everything inside them) travel in that slot, over its duration, instead of with  **obj** — "camera first, then the screw comes forward, then the insertion". Without the capsule they move with everything else. A following object can be spotlighted too: on that step its pose is the picture's, on every other step it rides its target.

The place is stored as fractions of the frame, not in millimetres or degrees of lens, so a different lens on the step or an export at another frame shows the same picture. The step's own saved camera is what counts: authoring against a view you have not saved is allowed, as everywhere in SBS, but the step will show the object from *its* camera — the status line reminds you to  Update camera. A raw part has no transform of its own: put it in a folder (or **Make transformable**) and spotlight that. Global Mode never carries a spotlighted pose to other steps.

Brand — the company standard in one file

Tools ▶ Brand... manages a `.sbsbrand` file: the header items and logo, the header default, text styles, shape styles, constant-title positions, pinned positions and shared crop masks of a company.  **Save brand...** writes it from a project that looks right (and links the project to it); on a linked project it offers to **update** the brand — the revision number goes up — or to save a new one.

 **Load / update from a brand...** brings a project to the standard. Definitions already linked to the brand are **updated in place**: their identity inside the project does not change, so every text box bound to a style, every constant title and every pinned item simply follows — a new logo or colour scheme reaches the whole project in one step. What the brand adds is added; what the brand no longer carries is kept and reported. The project's own definitions (not part of any brand) are never touched.

A project that was **not** born from the brand first goes through the **matching wizard**, one page per kind of definition (text styles, shape styles, constant-title positions, pinned positions, crop masks, header items). The brand's definitions are on the left; the project's own are orange chips. Drag each chip onto the brand definition it really is, or pick it from the chip's **Move to...** list. Several chips on one brand definition are **merged** into it: every text box, shape or image that used them follows, and definitions that referenced them are re-pointed. **Keep as the project's own** leaves a definition outside the brand for good; **Delete** accepts only definitions nothing uses. A same-name match is pre-filled; a merely similar one is offered as a **looks like** "... " → hint you click if you agree. Positions and masks are drawn on a small frame — the brand's mark in green, the project's in orange — so they can be matched by where they are rather than by their names. After the wizard the project is linked, and later brand updates are silent.

Everything is listed before and after in a preview, definitions you edited locally since the last load are flagged (one button keeps them), and the update is a single undo entry. Positions are scaled when the brand was authored at another frame size. The old `.sbsheader` setup files load here too.

The look of the printed document travels too. When the project has a document, the brand carries its look: the page layouts you drew, the default layout, the header and footer (your own design with its pictures, or the standard three cells), the watermark, the company name, and the numbering and contents

options. Loading the brand lists what would change in *this* project's document in the same preview — **Apply, but leave the document as it is** takes the rest without it. A page layout with the **same name** is replaced in place, so every page wearing it keeps wearing it; a layout the project lacks is added; the project's other layouts are left alone. The project is meant to look like the brand: a brand saved with the standard header puts the standard header back, and the preview says so first. Pages, texts, custom step numbers, extra pages, the title, the document number and the revision are **never** part of a brand. A header picture that shows a step of the animation stays behind too — it means nothing in another project. A project with no document yet gets an empty one wearing the look; no pages are made until you build them. The whole update, document included, is still one undo entry.

Everything a brand file holds (format 2):

Section	What travels	What stays with the project
Identity	Brand name, revision number, when it was saved, the export frame size it was authored at (positions are scaled into another frame size).	—
Header items	Every item of the film header: custom text (text, font, size, colour, effects), step name / step number / chapter number / chapter progress, subtitle line, pictures and the logo (the picture itself, embedded), each with its position and size.	Per-step hiding of a header, and the step numbers themselves.
Header default	The default text look for new header items (font, size, weight, italic, underline, colour, fill).	—
Text styles	Name, font family, size, bold / italic / underline, colour, box fill, drop shadow, outline.	Which text boxes are bound to which style, and their text.
Shape styles	Name, fill, outline colour, outline thickness.	Which shapes use them; corner radius (geometry).
Constant titles	Name, anchor corner, position, the text style they use.	Each title's words per step, and its width.
Pinned positions	Name, anchor corner, position.	The pictures and shapes pinned there.
Crop masks (shared)	Name and the rectangle, as fractions of the frame.	Private masks on single pictures.
Document look	Page layouts you drew (up to 40), the default layout, the standard header and footer cells, the designed header / footer bands with their items and the pictures those use (embedded), the watermark (text or picture, mode, tint), the company name, and the options: numbering, picture numbers, drop silent steps, contents page, contents with steps.	The pages themselves, the per-step text, title / document number / revision fields, a band item that shows a <i>step</i> of the animation.

Not in a brand, by design: CAD models and their colours, steps, overlays, narration, animation presets, camera templates, cables, notes, primitives — all of that is the project's content, not the company's look.

Brand files and older versions of SBS. A brand saved by this version is file format 2. An older SBS that knows about formats opens it, says that it comes from a newer version, loads what it understands and refuses to save over it — saving from there would drop the document's look from the file.

Changing a brand element inside a project. A text style, shape style, constant title, pinned position, crop mask or header item that came from the brand can still be edited in the project — the edit stays in *this* project. SBS tells you when that happens ( *Text style "Title" belongs to the brand "ACME"...*, once per element per session), because other projects will only get the change after **Tools ▸ Brand... ▸  Save brand...** updates the brand file. If you close the window with such changes still unsaved to the brand, SBS asks first: **Save the brand...** (the window stays open; save, then close again), **Close — keep them in this project only**, or **Cancel**. The document's look is watched the same way, part by part — a page layout that came from the brand, the default layout, the header and footer, the watermark, the company name, the numbering options ( *Document look "Watermark" belongs to the brand...*); a page layout of the project's own is nobody's business. Only changes made in the current session are asked about, and an undo that restores the element clears the question by itself. The unsaved-project question, if any, comes after it.

When a linked brand file has a newer revision on disk, SBS says so after the project opens, and the panel shows **↑ Update to the new revision**. Steps whose look changes get a ★ as with any other definition edit.

Hiding a header on some steps

Right-click a header element in the viewport ▸ hide on this step or on the selected steps. The hide list is stored on the step, so it travels with copies and imports. The eye in the Header tab hides an item globally instead.

Headers in export

Headers are composited after the 3D render, so editing a header never invalidates the 3D cache; only the stitching is redone. The subtitle item shows the current narration sentence, timed from the synthesised audio.

12. Narration, voices and subtitles

Writing the narration

Type in the field under the viewport. The text is stored on the step. Pasting from a document is sanitised (smart quotes, invisible characters and paragraph breaks that upset the voice engine are normalised). A change to the text starts the step for re-render because its audio and duration change.

Local voices

SBS ships a neural text-to-speech engine that runs on your GPU (WebGPU) with an automatic CPU fallback. The voice is chosen per language in **Edit ▶ Languages...** ( click to choose). The **Export** tab has **Voice** and **Speed** for the active language. Clips are cached under `audio/<lang>/<voice>/`; **Clear inactive voices** deletes the subfolders of voices you no longer use (safe — they re-synthesise on next play), **Clear all cache** wipes everything.

Slow or silent voice? Open Developer Tools (`Ctrl` + `Shift` + `I`), Console, and run `await window.sbsTTSDiag()`. It reports the engine, device, warm-up state and the last error in plain words. See chapter 21.

Cloud text-to-speech (optional)

Settings ▶ Cloud TTS accepts a Google Cloud Text-to-Speech API key. When set, cloud voices appear in the voice picker. Audio still lands in the same cache folder, so exports are offline once synthesised.

Subtitles

Add the  **Subtitle** item in the Header tab. Subtitles are split into sentences and timed from the audio. The narration of other languages comes from their language packs (chapter 13).

Title Manager (constant titles)

Edit ▶ Title Manager... manages the texts that repeat across steps:

-  **Constant titles:** each constant text box type, how many instances it has, and   to jump to the previous / next step using it. **Delete every type with zero instances** tidies up; the menu item **Clean Up Unused Constant Titles** does the same.
-  **Translated titles:** side-by-side original and translation per language; ✓ **Accept translation**,  **Update to translated version**,  marks rows changed since you last looked.

-  **Replace rules:** find / replace rules applied to titles and narration in one language.  **Bake** applies every rule and then deletes the rules — the corrected text becomes the text.

Edit ▶ Unify Constant Titles ( +  + ) merges text boxes that carry the same text into one constant, so a later edit changes them all.

13. Languages

One project, many languages. The project file keeps the **source language**. Every extra language is a sidecar file `<project>.<lang>.sbslang.json` holding that language's narration, header texts, overlay texts and constant titles, plus its own audio and render folders.

Edit ► Languages...

- **+ Add language** creates a pack (for example `he`) and translates the whole project into it with the engine chosen in **Settings ► Translation engine** (an OpenAI-compatible server URL and model name, with a glossary of terms that must not be translated; parallel requests configurable).
- Click a language to make it **active**: the viewport, narration field, header and exports switch to it. The status bar shows the active language.
-  chooses the voice for that language;  re-translates the parts that changed in the source;  removes the pack (the sidecar file is deleted after confirmation).
- **Settings ► Preferred narration languages** orders the languages the voice picker offers first.

What is per language and what is shared

Per language (in the pack)	Shared (in the project)
Narration text, subtitles, header texts, overlay text boxes, constant titles, replace rules, voice choice, audio cache, render cache	Model, tree, steps, cameras, colours, overlay geometry (positions, images, shapes), header layout, animation, hardware, cables, notes

Editing a text in a non-source language edits only that pack. Editing the source marks the corresponding rows in the other packs as stale ( in the Title Manager and in the Languages panel) so you can re-translate just those. Exports are made per language; the render cache is per language too, so a text-only change in Hebrew never re-renders the English video.

Translation and proofing sheets (Excel)

For a translator or a proofreader who works outside SBS, **Tools ► Review form...** opens a panel with  **Create sheet...**,  **Import sheet...** and the list of review notes. Both directions work while the project shows its original language.

Export writes one `.xlsx` workbook into the project's `exports/` folder (you can choose another place) with **one tab per language** you tick — each translation language, and optionally the original as a proofing sheet whose Target column starts empty. One row per line of text, in timeline order: a **Chapter** row wherever a chapter starts, then each step's **Voiceover**, **Step name**, **Title** and **Text box** lines, and the project **Header** texts at the end. Every step gets a Voiceover row even when it has none yet, so the client can write one. The **Step** column shows the number exactly as the Step Number header shows it (per-

chapter restart honoured, sub-steps as 5.1, 5.2). Hidden steps and hidden chapters are left out unless you tick **Include hidden steps**. **Source** holds the original text, **Target** the current translation, **Notes** is free for remarks back to you. Optionally a **Preview** picture of every step sits next to its voiceover — freshly rendered at 300 px (SBS walks through the steps once) or the small step thumbnails. Rows are colour-banded so a step reads as one block: chapter rows blue, steps alternating two tones, a line where each step starts. The visible **key** column is the step's permanent identity (locked; a hidden fingerprint column sits next to it). The sheet is protected: only Target and Notes accept typing; rows may still be sorted and filtered.

Import accepts `.xlsx` and OpenDocument `.ods` (what Apache OpenOffice and LibreOffice save natively; the old binary `.xls` is not read — choose "ODF Spreadsheet" or "Excel 2007+" when saving). It reads every language tab of a returned workbook, one after the other, and matches every row by the hidden **key** column — the step's permanent identity, so it does not matter that steps were added, moved or renumbered while the client worked. If the key column was deleted, rows are matched by the source fingerprint, then by exact source text; a row that matches nothing is reported and skipped. A row whose source text you edited after the export is flagged  "source changed after the export", and one button applies only the rows that are still safe. Empty Target cells never erase an existing translation. A sheet from another project asks before proceeding. Everything is listed before and after in a preview, lands as one undo entry per tab, and — for a target language — goes into its pack; switch to that language to see it. A proofing sheet of the original changes the project text itself, and the other languages then show those lines as drifted.

Review notes (tab)

Whatever the client typed into **Notes** — and any picture they floated over a row, a screenshot for example — comes back as a **review note** tied to that step. A step's several rows (voiceover, name, titles) fold into one note, each cell labelled by its row type, pictures collected. The  tab next to **Export** lists them: a tick marks a note as addressed, the filter shows open, addressed or all, and clicking a note jumps to its step and opens the note window.  **Open note window** opens that movable window on its own: full text, the pictures,   through the list, **Addressed**, **Go to step**, delete. Notes are saved in the project;  **Clear addressed** removes the ones you have dealt with.

Screenshot slot — drop a PNG at the path shown below.

Screenshot: an exported sheet open in Excel — chapter rows, step numbers, previews, Source and Target. File: `docs/manual/img/translation-sheet.png`

14. Hardware: screws, nuts, washers

CAD assemblies often ship without fasteners, or with hundreds of identical ones. The  tab lets you define a screw once and drop instances onto faces.

Templates

+ **Create template** opens the editor: **Head type**, **Drive style**, **Ø Diameter (mm)**, **Length (mm)**, **Thickness**, **Colour**, washer names, **Spacing X (mm)** and **Gap scale** for stacked washers, **Reposition (ms)** for the pre-insertion move. Save with **Save changes**.  edits,  deletes,  restores the defaults from **Settings ▸ Hardware insertion defaults**. Editing a template changes every instance made from it (the tree menu says so:  **Edit template (affects all instances)...**).

Placing

Click **Assign** on a template, then click a face on the model: the screw is placed normal to that face, head outside.  or right-click cancels. In the tree, right-click a screw for  **Duplicate (same template)**,  **Add nut** /  **Delete nut**, the washer stack ( **No washers ...**  **Spring + flat washer**) and  **Adjust insertion animation....** The **HW** filter in the tree shows only hardware.

Insertion animation

When a screw is hidden in step n and visible in step n+1, SBS animates it driving in: it first appears at the reposition distance along its axis (over **Reposition (ms)**), then travels in over the **insert** channel. Nuts and washers follow the same rule in reverse order. Hide a screw in a later step and it backs out.

15. Cables and sockets

A cable is a smooth tube between two sockets with a controllable path. Sockets are small blocks you attach to parts; cables plug into them. Both are animated between steps.

Creating

 tab ▶ + **New Cable**: name, **Cable color**, **Diameter (mm)** (the tab's diameter field sets the default for new cables), socket **W / H / D (mm)** and **Socket color**. Place each end by clicking a face (the placement icon on the row), then add mid-points by clicking along the path. Duplicate a cable with **Duplicate** and choose **Keep connections** or **Clean connections**.

Cables in steps

Each step stores the cable's point positions and which socket each end is plugged into, so a cable can be unplugged in step 12, dangle in step 13 and be plugged into another socket in step 14. Editing points in a step changes only that step; the eye and highlight toggles on the row are per step too. When the part a socket is attached to moves, the socket and the cable end follow. A step's context menu lists the recorded cable actions (**cable · plug/unplug · socket · State n**).

Sizes are absolute millimetres. Projects made with the older percentage sizing are converted on load so they look the same.

16. Flat shapes and primitives

Flat shapes (■)

A flat shape is a filled outline traced from a model face: highlight a gasket surface, mark a sealing area, show a footprint. Click the trace button, then a face — adjacent triangles within **Crease angle (degrees)** of the picked triangle's normal join the face set and their outline becomes the shape. Tighter = only flat regions; wider = catches gently curved surfaces. Shapes belong to **shape groups** (double-click to rename); the eye on a group toggles every instance in it; the lock makes the group select as one.

- **Edit** re-opens the polygon in the viewport editor seeded at an existing instance.  **Edit shape...** in the tree does the same — and so does a **double-click** on the shape in the viewport. While editing: double-click a corner to delete it, an edge to add a corner there, inside an outline to take the whole outline (its move / turn / scale handles).
- **Style**: fill colour and opacity, outline colour and thickness — per shape, or from a shape style.
- Textured plane: pick an image file, then click a face (or empty space) to drop a 2D plane sized to the image's aspect ratio — decals, labels, screens.
- Shapes appear, move and fade with the **shape** channel and keep their end state exactly as the step says, even when the part under them fades out.

Primitives (⬡)

Boxes, cylinders, spheres and similar helpers built inside SBS, used as stand-ins (a missing bolt, a tool, a fluid volume) or as containers. Choose a type and size in the ⬡ tab, then click a face to place it tangent ( **Esc** cancels), or  **Align to surface...** / **Align by 3 points** later. Primitives have a base face at the origin so they sit on surfaces naturally, and they can hold children like folders: drop parts inside and they move with it. Right-click a primitive:  **Paste (independent)** versus  **Paste Instance (linked parameters)** decides whether a copy shares its dimensions with the original. Sizes and quality settings live in the project and are part of what the render cache fingerprints.

 **Editable poly (Poly box)** — a box you model, the way an editable poly works in 3ds Max. Create a **Poly box** from the ⬡ palette (or right-click any box ▶  **Convert to editable poly**), then right-click it ▶  **Edit poly...** — or, with the poly selected, press  (vertices) or  (faces) to go straight in, as in 3ds Max ( is kept for edges). In **faces** mode (): hover highlights a face, click selects it,  or  +click adds or removes, a drag that starts off the poly box-selects ( adds,  removes,  = only faces fully inside); drag moves the selected faces in the view plane, or use the **gizmo** that appears at the selection's centre to move / rotate along axes ( + an arrow extrudes along it) —  (or the badge) cycles its axes between **FACE** (the average normal), **WORLD** and **PARENT** (the object's own frame), and typing a number during a gizmo drag applies that exact amount (maths allowed,  keeps,  cancels).  +drag **extrudes** the selection along its normal — type the distance while dragging for an exact value — and on release the shape is **joined** (pulled out: the block unites with whatever it runs into, as in

3ds Max; pushed in: it cuts a pocket, or a through-hole when it reaches the far side). The edges you made stay: a loop cut survives a join;  **Clean edges** (right-click the poly) merges coplanar faces when you want the simple shape. Hold  over an edge to **preview** the loop cut,  + click to cut — the loop runs through quads at the same fraction and through any other face straight across. In **vertices** mode () drag a dot ( /  adds more) or use the gizmo. Every gesture is one undo, and undo / redo work while the mode is open;  cancels the gesture in flight, otherwise ends the mode, as does clicking off the poly, changing step or starting an export. The shape is the primitive's own parameters, so it saves with the project, shows in every step and takes a colour preset like any primitive;  **Export poly as .glb...** writes it as a file (with the topology inside, for a later re-open). Not yet: edge mode, deleting faces, projecting a picture onto it.

 **Poly Editor** — a bunch of objects go in, one asset comes out. Select anything in the scene (primitives, parts of an imported model, folders, editable polys), right-click ►  **Edit in Poly Editor...**: the app first asks for the object's **name** ( takes the suggested one) — it is the name of the folder everything goes into, even a single part, and the name the .glb is saved under. Every object becomes an editable poly — each stays its own object — and the editor opens over the same viewport with the project hidden. The folder's pivot is the middle of what it holds: the folder sits in the world, and what is in it is measured from the folder (the gizmo's *Parent*). On the left is the editor's **own tree** (folders and parts): drag rows to arrange it, double-click to rename,  **Folder** wraps the selection. **Add** puts a primitive (box, cylinder, sphere, cone, pyramid, tube, torus, capsule, geosphere) where you are looking, sized to the view: it is an editable poly from the first moment, with few enough sides to model with. The panel has four tabs:  **Model** (what is described here),  **Refs** — **reference pictures** to model against: go to a flat view (Front, Back, Left, Right, Top, Bottom) and  **Add a picture**; a photo is first *squared up* — drag four round corners onto something that should be a rectangle and it is re-formed so that it is one (a drawing can be used as it is). What you can trust in the photo is often smaller than what you need: the square handles on the four edges pull the **frame** out (or in) while the perspective the corners fixed stays, so the frame can take in the whole object;  **Frame = the corners** puts it back. The top-left corner of the viewport carries the editor's views (Persp, Top, Front, Left, Right, Bottom, Back),  **Fit** and  **Pictures** (move / scale the reference pictures) — the project's work camera () is not used in the editor. A selected part always shows its edges, drawn over its texture and over the reference pictures (also the ones set over the model); in Vertices / Faces the edges and the selection draw over them too, and the projected texture stays on. In that dialog the mouse wheel zooms in and out at the cursor and dragging the picture (or the middle button) moves around it, to place a corner exactly; double-click or  **Fit** shows it whole again. A picture on a view can be stretched wider or taller than its own proportion:  + pull a corner in  **Move / scale** (without Ctrl a corner keeps the proportion);  **Proportion** under the selected picture puts it back. PNG pictures keep their transparency, also when they are projected. **Projection** (Refs tab ►  **Project**, or right-click a part): the pictures are projected onto the selected parts (none selected = all of them) as a box projection — every face takes the picture of the side it faces most (right / left, top / bottom, front / back), straight along that axis. If a face looks toward a side that has no picture while the opposite side has one, the app asks whether to project that picture *through* the part onto it (a mirror image) — per side. Move, scale or square a picture and its projection follows;  **Remove** takes it off. All the pictures of one view are **merged** into one before they are projected, the way they lie on the view — the higher in the list covers the lower, like layers: square up a close-up of a far part, enlarge it over the whole shot, add a sticker on top. The  on a picture's row leaves it in or out of the projection. Outside the pictures a face keeps the part's own colour.

 **Bake** (or right-click ▶ Bake the projection) freezes the projection into the part's **own texture**: every face gets its own spot in one picture, painted with what the projection puts there over the part's colour. From then on the texture belongs to the part — it moves, turns and is copied with it, so one textured object can be duplicated into many — and  **Project** again starts a fresh, live projection. Apply bakes whatever is still live and writes the textures into the .glb (texture coordinates and the picture inside the file); in the project a textured part wears a white colour (a colour tints a texture).  **Remove** takes a projection or a baked texture off; a part whose faces change after baking (extrude, cut...) shows its colour until it is baked again. The picture stands in the scene on the object's centre (the pivot of its folder), facing its view: it grows and shrinks with the zoom and shows only in that view, never in perspective.  **Move / scale**: drag a picture to move it, pull a corner to scale it — where you leave it is its home; each picture also has an opacity, a width, *over the model* (to trace on top of it), hide, order in the stack, square up again, remove. The pictures are kept with the asset by their file's path —  **Colours** — the project's own Colours panel, the same one as the Colors tab: the list, the edit card (colour, solidness, metalness, roughness, reflection, outline...), + **Add Color**, rename, delete, unify and the right-click menu all work as they do there, on the project's colours — an edit here changes the colour in the project too. **Assign to Selected** puts the open colour on the selected parts of the editor, and selecting parts marks the colours they wear. Colour, solidness, metalness and roughness show in the editor; outline and reflections show after Apply. There is no "default" in the editor (its parts are not project parts yet). After Apply the parts wear these same colours in the scene (no look-alike copies) — and  **Environment** — a background colour for the editor only: it is for seeing things there, the project's background is not touched and comes back when the editor closes. To build an object that is not in the scene at all, open the editor empty: Primitives tab ▶  **New asset in the Poly Editor....** How the tree is arranged is how the objects are separated in the saved asset. In place of steps there are **views** (Persp, Top, Front, Left, Right, Bottom, Back — the axis views are flat; the editor uses a hundred times longer lens there than the project's "orthographic", so vertices far behind line up with the ones in front; at that distance close surfaces can flicker against each other) and  **Fit** (). Click a part to select it ( adds); the gizmo moves / rotates whole parts. Every part and every folder has its own **pivot and axes**, and the gizmo works in one of three sets of axes ( cycles them): **World**; **Parent** — the axes of the folder it sits in (the world, at the top level); **Local** — the object's own axes. Inside a part (vertices / faces) the same three mean: world · the selected faces' normal · the part itself. **Right-click the gizmo** for a small panel: the three buttons, and the object's *Position* and *Rotation* in the chosen space — in World they are measured from the world's origin — when you are editing an asset that is already in the project, the editor's world is *the asset's own* origin and axes, wherever the step has put the model — in Parent from the pivot and axes of its folder (no folder: the world), and they follow the object as it moves. Type a value and press  to put it there: 0, 0, 0 brings it back onto that origin. In Local the fields always read 0 (an object is at its own origin), and a typed value moves / turns it *by* that amount along its own axes. A folder's pivot is a fixed point (the middle of what it wrapped when it was made), so what sits in it keeps a steady position. **Right-click a part** (or its row in the tree):  **Align to a surface...** — first click a face *of the object* (the face that should touch), then the face of another part it should sit on: the two faces meet at the clicked points;  **Align by 3 points...** — three points on the object (three corners of the face that will touch, or three points of a rim), then three where it goes: the two planes meet face to face — the object sits against the other surface, not inside it — and the centres of the two circles coincide;  **Pivot** — *Move the pivot only* (the gizmo then moves the pivot and the geometry stays;  ends it), pivot to a surface point, by 3 points, to the centre, to the base, or its axes back to the world's. While one of these

tools waits for a click, a pin follows the cursor — a cross with an arrow along the surface — and shows what the click would take; each picked point keeps its pin (cyan on the object, orange where it goes). A click snaps to a **pivot** (of a part or a folder — drawn as small diamonds while picking), to a **corner**, or to the nearest point on an **edge**; otherwise it is the point on the surface. **Scale** (panel ▶ ↗ **Scale**, or right-click ▶ Scale): a 3D box stands around the selection — one object: along its own axes; several: along the asset's. Each of its six faces carries a **pyramid**, and each face that looks at you a flat **triangle** on every corner. Pull a pyramid to stretch the box in that direction: the opposite face stays; hold **Alt** for both sides at once (from the centre), **Shift** to scale every direction equally. Pull a corner triangle to scale that face's two directions like a standard scaler: the opposite corner stays and the third direction is untouched; **Shift** keeps the proportions it had, **Alt** scales from the centre, both together do both. The very tip of each corner is **white**: pull it to scale everything by one amount — width, height and depth together — with the corner straight across the box staying where it is (**Alt**: from the centre). **Esc** ends it; one undo step per pull. With one object selected the panel shows its **Size %** along its own X / Y / Z — 100 % is how it came into the editor; type a number to rescale it about its pivot, and 100 / 100 / 100 (or 0) gives back the proportions it came with (exact while it was scaled along its own axes; an object that was stretched at an angle, inside a turned folder or a group, comes back approximately). **Shapes** (Model tab ▶ Shapes) — the project's flat-shape tool inside the editor: ➡ **New shape on a face** (click a face of a part — or empty space — then the corners; click the first corner, right-click or **Enter** closes it), ➡ **New shape on the view** (in a flat view: drawn on that view through the object's centre, whatever geometry is there) and □ **Shape from a face** (a face and its neighbours within the angle). A shape is a flat part in its own frame — the plane it was drawn on — so turning it turns its system. Double-click it to edit its outline: drag a corner, double-click a corner to delete it, an edge to add a corner, inside to take the whole outline (move / turn / scale); right-click adds an outline (XOR) or the outline of a face. To make it a solid: Faces (4), click its face, **Shift** + drag the gizmo. **Booleans** (right-click a part ▶ ⊕ **Boolean**, or the Union / Subtract / Intersect buttons under Tools): select the part to keep, choose the kind, then click the other object (with two or more already selected, those are used — the primary is kept). The result is shown in their place before anything changes: ✓ **Apply** (**Enter**) bakes it — the kept part takes the result, the other parts leave the tree — and ✕ **Cancel** (**Esc**) leaves everything as it was; the bar can still change the kind, and ⇄ **Swap** turns a subtraction round when the wrong object was kept. No record of the operation is kept; one **Ctrl** + **Z** takes it back. Both objects must be closed solids (a flat shape is not — extrude it first). **Restore** (right-click ▶ ↶ **Restore**): the selected parts go back to how they came in — shape, pivot, colour; for an asset being re-edited that is the saved file — while the rest of the edit stays; it is one undo step. Pivots and axes are saved inside the asset, for the next edit; 1 / 4 or a double-click go into the part's vertices / faces — the same tools as a Poly box, joins and cuts included; **Del** deletes, **Ctrl** + **D** duplicates, **Ctrl** + **Z** undoes inside the editor only. Flat faces come in as polygons; a curved CAD surface stays triangles (it is the model's own tessellation). ✓ **Apply** is the only way out with a result: the tree is written as one .glb into the project's models folder and loaded straight back as a model whose tree is the one you designed. In the scene, a selected part that belongs to such an asset is highlighted in **orange** instead of the usual selection colour — the sign that right-click ▶ Edit in Poly Editor will open that asset. **Editing it again:** right-click that model — or any part inside it — ▶ □ **Edit in Poly Editor...**: the whole asset opens (it is one file), polygons kept. ✓ **Apply...** always asks how to save. **Replace <file>** (the default) saves the edit over the same file and brings the same model up to date in every step — a straight swap, nothing else is asked. The whole asset comes

in as the file has it — also the parts this step has hidden, archived or dragged out of the model. A part you only reshaped keeps everything the steps know about it (visibility, colour, what is attached, the folder you dragged it to); a new part appears in every step; a part you moved to another folder follows that folder wherever it still sat in the old one. A removed part leaves every step — unless something is still attached to it (a note, a shape, a screw, a cable end), in which case nothing is written and the app says what is attached. The previous file goes to the project's **backups** folder, and the app also keeps it in memory: the undo history from before the Replace is cleared, but the Replace itself can be undone — **Ctrl + Z** writes the previous file back, returns its parts to the scene and puts every step back exactly as it was (**Ctrl + Y** replaces it again). That holds until models or steps are added or removed in the project: after that the Replace can no longer be undone there, and the file in **backups** is what is left. If you changed the tree (added, removed or moved parts) the question says so: other projects that use the asset get the new tree when they open; with the tree unchanged they simply get the new shapes. A project that opens and finds the asset changed says so in the status line and stars ★ the steps whose picture changed; what hung on a part that no longer exists moves up to its parent. **Save as a new asset...** asks only for a name: the new .glb is added to the scene as a separate model, where it stands, and the old model is left exactly as it is. Nothing follows and nothing is removed — to let the new one take the old one's place (Files tab ► Browse on the old model), to make it follow an object, or to archive / delete the old model, do that in the project afterwards. A new asset made from objects of the project: Apply asks for its name, then the app asks what to do with the originals — archive them (recommended), keep them, or delete the ones that can be deleted (parts of imported models can only be archived) — and can put them in a selection group (Select tab). **✕ Discard** leaves the project exactly as it was.

17. Notes

A note is a callout attached to a part: a label with a leader line that follows the part through steps and camera moves. Right-click a part ▶  **Add Note...**, type the text, pick a size (● **Size: Small / Medium / Large**) and a template from the  tab.  **Reposition Note...** lets you click a new anchor;  **Delete Note** removes it. Notes are per step: the part's note list is stored in the snapshot, so a note can exist in steps 5–8 only. The  tab keeps **note templates** (style, size, leader look) and the list of notes in the active step. Note text is translated with the language packs.

18. Environment and look

Env tab

- **Theme:** **Light** / **Dark** for the interface; **Default background** in Settings for the viewport colour of new projects, and the background picker here for this project.
- **Grid** on/off, **Outline** on/off with **Outline color** and the **Small** / **Medium** / **Large (px)** thickness presets. Outlines are a silhouette pass around parts; per-preset override in the Colors tab.
- **Lights:** **Key light** (warm), **Fill light** (cool), **Rim / backlight** — intensity and angle. The light stage window lets you drag them around the model.
- **Load an environment from a .sbsenv file** — or **grab it straight from another .sbsproj**: copy a whole look between projects.

Production render settings

The render settings panel (from the Env tab) controls what the export looks like: studio presets (**Built-in studio** (default), **Smooth gradient** — **no features** (a featureless gradient sphere: nothing in it can draw an edge on a flat part; also in the Light Stage cycle), **Studio** — **bright**, **Studio** — **soft boxes**, **Photo studio** — **warm / earthy**), **Environment strength** and **blur**, **Exposure**, **Contrast** and **Saturation (grade)**, ambient occlusion (**Intensity**, **Radius (px)**, **Steps**, **Thickness**, **Distance falloff**, **Max distance**) and the rim light spread. Changing any of these changes every frame, so it stars all steps.

Preview versus export. The viewport shows the same renderer as the export, at the viewport's size. Exports add supersampling per frame, so edges are cleaner than on screen.

19. Export, render cache and stars

Export tab

Field	Meaning
From / To	Step range to export. Leave empty for the whole project.
Width / Height (px), Format preset	Output size. Cache clips are rendered at this size; changing it re-renders everything.
Frame rate (fps)	Typically 30. Part of the cache key.
Step hold (ms)	Extra dwell appended to each step after its narration. A pause <i>within</i> a transition is authored in the animation preset (pause channel).
Voice · Speed	Narration voice and rate for the active language.
Output format · Quality	MP4 (H.264) or WebM; the quality slider is the encoder bitrate cap (variable bitrate — static content uses less).
File name · Choose folder...	Defaults to <code>exports/</code> inside the project.
Alpha masks	Also write a per-frame alpha (matte) alongside, for compositing in another tool. Off by default.
.sbsproc	Export a single self-contained file (MP4 + step manifest) for the SBS viewer.
Start Export · Cancel Export	Run / abort. Progress and the current step are shown in the status bar.

Settings ▶ **Export defaults** sets the defaults for new projects (output format, frame rate, step hold).

How the render cache works

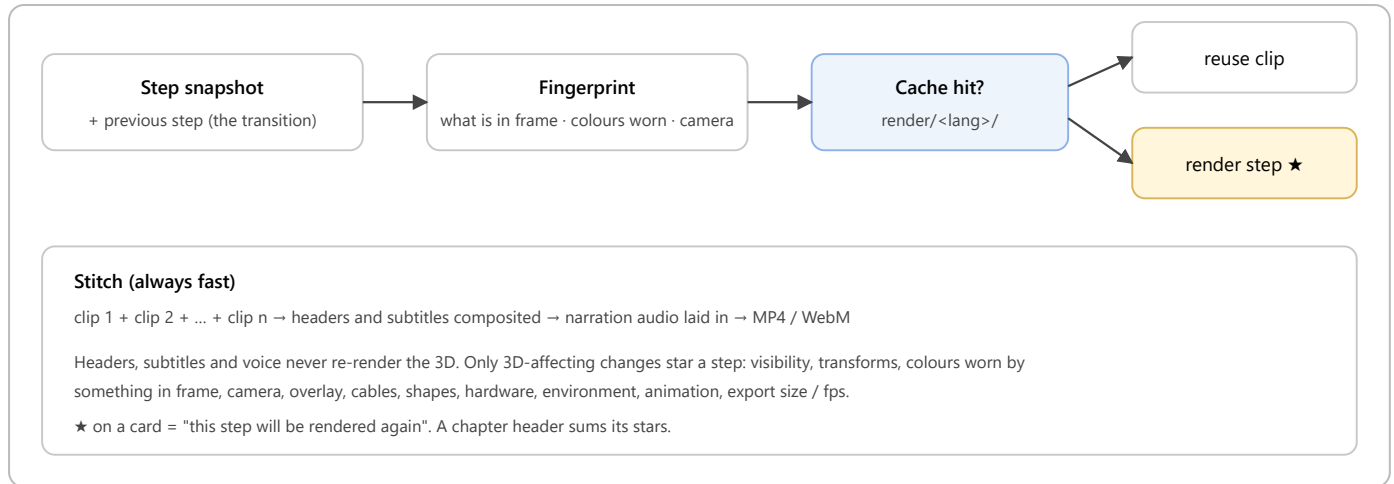


Figure 6 — Each step is a clip in the cache. Export = render the starred clips, then stitch everything.

The first export renders every step. From then on, **Start Export** opens a small prompt:

- ★ **Re-render starred** — render the starred steps, reuse the rest. The normal choice.
- **Trust the stars** — if the cache keys changed for a technical reason (an app update changed the fingerprint format) but nothing in your project did, adopt the existing clips instead of rendering. Use it once after such an update; if a clip then looks stale, star the step and re-render. When the **render settings** differ from the last render's (an update added a render option, you changed the background or the quality...) you are **asked**, with the list of what changed: OK reuses last time's clips and renders only the starred steps, Cancel renders everything. Only a change of size, frame rate or cache generation refuses on its own — clips of two sizes cannot be joined.
- + **current** — also re-render the active step even if it is not starred.
- ► **Render full project (incremental)** and 🔄 **Re-render selection** with a range like 0-10, C11, 124, 89-102 (C11 = chapter 11). With **When finished, assemble the complete video unticked**, the selection renders into the cache and nothing else is touched — segments outside it that are out of date are left alone and counted in the status line. Ticked, the whole video is assembled: out-of-date segments outside the selection render too, unless **Trust the stars** says to reuse them.
- 🧹 **Purge stale cache** — delete clips no step refers to any more (old sizes, deleted steps). Never deletes clips that are in use.

Stars: what changes what

A step gets a ★ when something that affects *its rendered frames* changed since it was last rendered. SBS compares against the rendered version, so undoing the change removes the star again. The rules:

You change...	Stars...
Visibility, a transform, a colour assignment, the camera, the overlay, narration text, hold, easing, private animation of step n	Step n, and step n+1 (its transition starts from n).
A colour preset's values	Every step where a part wearing that preset is in frame (see the frame scan below). Presets nobody wears in frame star nothing.
A text style, shape style, linked shape, constant title, pinned position, crop mask	Every step whose overlay uses it.
A camera template	Every step bound to it.
A cable's look (colour, diameter, style), a primitive's size or quality, a flat shape	Every step where it is visible.
Step order, chapter membership	The steps whose predecessor changed.
Animation preset (project default), AL1 / AL2 defaults, environment, export size, fps	All steps.
Header layout, subtitles, voice, narration speed, language texts	Nothing — composited at stitch time.

Deliberately not starred. Editing a hardware template, a note template, or the geometry of a loaded model (relink, source transform) does not star steps automatically. After such edits, right-click ►  **Re-render cache segment** on the affected steps, or clear the cache.

Cleared the star by mistake? ☆ **Clear star** only tells SBS to keep the current clip. If the clip looks wrong, right-click ►  **Re-render cache segment**.

Scan steps for what is in frame

Edit ► Scan steps for what is in frame... walks through every step with its own camera, at the export aspect ratio, and records which parts actually appear in the picture (ghosted and transparent parts count as visible). The record is saved next to the render cache and has two uses:

- Stars become precise: changing a colour preset stars only steps where a part wearing it is really on screen, not merely un-hidden somewhere behind the camera.
- Cache keys become narrower: a change to something off-screen no longer invalidates the step's clip.

The scan takes a moment per step (it is a GPU pass) and reports progress in the status bar. Records are refreshed automatically for steps you leave after editing them and at the end of every export; run the menu command after large reorganisations or when you want the stars to settle before a big export.

Exporting other languages

Make the language active in **Edit ▶ Languages...**, then export. Each language has its own render cache folder, so switching languages does not throw away work; only steps whose overlay text differs are rendered per language.

Document — the animation as a printed manual (PDF)

Tools ▶ Document... turns the approved animation into a paged A4 document: every page covers one or more steps, with their text on top and a picture of their final state below, inside a fixed page template with a header and a footer. The document lives inside the project file — there is nothing separate to keep in sync by hand — and it keeps its own copy of everything: **nothing you do in the document changes the animation**, its steps, its step groups, its numbering or its voiceover.

The document workspace

The document opens as a workspace that takes over the whole window; the animation's shortcuts are silent while it is open. Ctrl+S still saves. Ctrl+Z / Ctrl+Y work on **document edits only**: when the next thing to undo is an edit you made to the animation before opening the workspace, undo stops and says so, rather than changing the animation out of sight. ◀ **Edit animation** (top-right) closes it — nothing is lost; on the animation side the same corner holds **Edit document ▶** (above the steps panel), so the two buttons work like a toggle between the animation and its document. **Drag the separators** between the three panes to give the page more or less room (the widths are remembered on this machine); double-click a separator to fold that pane away and again to bring it back. On a small screen the page list gives way first and the settings fold away by themselves, so the page always stays usable.

Where	What
Right	The document page by page , under its chapters. Every box shows a thumbnail of the page itself — header, text, pictures — with its printed page number. A joined page is one box with one thumbnail, labelled <i>3 steps merged (steps 2–4)</i> , and lists its steps under it as compact rows; each step row ends with an eye that leaves the step out of the document. The contents and custom pages carry a ⋮ grip: they are the only things that can be moved. Right-click for the page commands. The list scrolls; ↑ / ↓ walk the steps, ← / → (or PageUp / PageDown) walk everything that prints — the same arrows that step through the animation.
Centre	The selected step's page — A4 in true proportion, header and footer included. It is the very page the PDF is printed from, so what you see is what prints. Fit / 100% set the zoom.
Left	The document's fields (title, company, document number, revision, numbering) and the selected page: its template, its pictures, its ! marks and the state of each of its text lines.

Pages

- **Build the pages — one per step** is the starting point. A step with sub-steps counts as one unit (it plays as one clip); its sub-steps become extra lines on the same page. Hidden steps and hidden

chapters are left out.

- **Selecting.** A selection is always **one unbroken range**: click a step, then Shift-click or Ctrl-click the other end — everything in between is selected. There is no way to leave a hole in a range, so a join can never leave a step outside.
- **Joining.** With a range selected press ↵ **Join steps 25–28 into one page** above the list, or right-click the selection and pick the same command. The joined steps keep **the animation's own numbers** — 25, 26, 27, 28, not 25.1, 25.2 — because the manual and the animation must agree on what "step 27" is.
- **Splitting.** Select one step inside a joined page: ⌘ **Start a new page at step 27** (button or right-click). **Un-merge — one page per step** undoes a join completely.
- **Leaving a step out.** Click the 👁 at the end of a step's row (or right-click ▶ **Leave ... out of the document**; with a range selected it applies to the whole range). The row stays in the list, dimmed, with a shut eye 🙈; click it again to put the step back. This is a document-only decision: the step is **not** hidden in the animation, the other steps keep their numbers, and a page whose steps are all left out is simply not printed.
- **The layout follows the number of steps.** A page is on *Automatic* until you choose otherwise: join two steps and the page gets two pictures, three steps → three, four or more → four (sort the rest out by hand), always in step order; un-merge and the pages go back to one picture. Pick a template yourself and the page keeps it — choose **Automatic** again in the template list to hand it back.
- Each page has a **template**: *One picture — wide (16:9)*, *One picture — tall (3:2)*, *Two pictures side by side*, *Three pictures — one large, two below*, *Four pictures — two by two*, or *Large picture, short text*. Templates are strict on purpose: a picture can only live in a template frame, so every page of the manual lines up.
- **A step that holds a video** is pictured at one frame of that clip. The picture's bar then carries ✂ **Frame...**: scrub the clip's window, step a frame at a time with ◀ ▶ or the arrow keys, or type the exact second. Each picture keeps its own frame, so two frames of the same clip can sit side by side on one page — put the same step into two frames and pick a different moment in each. Without a choice, a video step shows the first frame of its trimmed window, as before.

Pictures are cropped into their frames. A frame does not need the video's proportions: the picture fills it and what does not fit is cropped. **Click a picture** to select it — then **drag** to move it behind its frame and use the **mouse wheel** to scale it about the pointer (one undo step per drag). The bar above the picture has **Fill** (back to the default crop), **Whole** (the entire picture inside the frame), **- / +** (5% a press) and a **percentage** you can type into — **100% fills the frame**, and anything from 10% to 800% is allowed. The wheel moves in 2% steps; hold Ctrl for 0.5% and Shift for 10%. If you make the picture smaller than its frame the white page shows around it — that is allowed. Esc deselects.

- **Step numbers on the pictures.** On a page that prints several steps every picture carries the number of the step it shows (the same number as its line), so the reader can match text and picture. Switch it off with **Step number on each picture** in the left pane.
- **Before and after.** A step picture normally shows the step's *end*. When the end hides what happened — a part that is already inserted — add the **before** frame: **Picture ▼ ▶** **before step 27**. It shows the state the step *starts from* (the previous step's end) seen through *this* step's camera, so the pair is

framed identically; it carries no overlay or notes (those describe a step's end) and is labelled 27 *before*. Put the before-frame and the step's own picture side by side in a two-picture page.

- **Which picture.** By default every frame is *Automatic*: the page's steps are shared out so that the **last frame always shows the page's final state** — four steps on a two-picture page show steps 3 and 4, on a four-picture page steps 1 to 4. **Picture ▾** on the picture's bar — or the **Picture 1 / 2 / ...** button of that frame in the left pane — opens a **picker with thumbnails**: every step of the page with its picture, number and name, under each one its *before* frame (tagged *before*; until it has been rendered the thumbnail is the previous step's, which shows the same state), and the current choice is marked. It lets you choose a specific step of the page, leave the frame empty, or take an **external image** — a photo or a drawing that is not part of the animation. An external image is scaled down to at most 2400 px and stored inside the project. A step picture is the step's export frame exactly as the video shows it — the step's recorded camera (even while Work Camera is on), 3D, overlay, notes and hardware tags — but **without the animation's header items and subtitles** (the page has its own header and its own text), and without selection highlights or other authoring marks. Step pictures are rendered when a page is shown or exported (the app walks the needed steps behind the workspace and returns to where you were); they are not stored in the project, so the file does not grow. A picture refreshes by itself when its step, its overlay, a colour or a style changes; after anything else (a reloaded model, render settings) use **Render the pictures again** in the left pane.
- **A step group lays itself out.** A group is one instruction made of several sub-steps, and its page gets **one picture for each sub-step that has a sentence** — two, three or four frames, picked automatically — with every frame showing the state *its own* sentence describes and wearing that sentence's number (6.1, 6.2, 6.3...). A sub-step with nothing to say gets no picture, because it has no line to belong to; a group where nobody speaks keeps its single end picture. A group with **more than four** speaking sub-steps is split into parts of up to four, as evenly as they go (five becomes 3 + 2, seven becomes 4 + 3), each part on its own page, so every picture is shown — and the numbers simply carry on across the break. In the step list a later part appears as a row of its own: merge it into the page before it if you would rather have the whole group on one page, or hide it on its own. Hiding the group hides all of its parts. Used deliberately, groups become a way of arranging the manual: group the steps that belong on one page, and the page arranges itself. A layout you picked by hand is never changed; press **Sync** to bring an older document's automatic pages up to date.
- **Numbers** in front of each line: the same step numbers as the animation (default), or 1, 2, 3 on every page, or none.
- **A step with nothing to say is not printed.** A step that only moves the camera or slides a part into place has no text, and a numbered line followed by white paper helps nobody: it gets no line and no number, the lines below close up, and the contents leaves it out too. Its **pictures stay** — a silent step can still be worth seeing. With 1, 2, 3 numbering the count simply carries on; with the animation's own numbers the speaking steps keep them (so a manual may go 10, 12 — the same numbers the film shows). It is gone in the editor too, not just in print. To see the empty steps again, untick **Leave out steps that have no text** in the left pane: each one comes back as an ordinary numbered line — which is also where you type to give one a sentence. To leave a step out completely, pictures and all, use the eye in the step list.
- **Your own number on a line.** Double-click a line's number badge and type what it should say — **Enter** keeps it, **Esc** puts it back. The **picture that belongs to that step shows the same number**, and so does the contents. It is a label for that one line only: nothing else is renumbered,

and clearing it returns the line to the automatic count. In the editor a typed number wears an orange ring so it can be told from the automatic ones; the ring is never printed.

Chapters and the table of contents

Every page shows the name of its chapter at the top of its text, so a reader who opens the manual anywhere knows where he is. On the chapter's first page the name is a real heading — that is what the PDF's bookmark points at — and on the following pages it looks the same without creating more bookmarks: one bookmark per chapter, however many pages it has.

Pictures of a step that shows an interface can be framed on the interface itself: tick  **Frame the picture on the interface** in the document settings and every such picture is zoomed so the panel fills its frame. Move or scale one yourself and **your** framing is kept from then on — switching the option off and on again never takes it back. The picture's bar then offers  **Standard framing** (follow the standard again) and  **Set as standard**, which makes the framing you just made the standard for **every** interface in the document — stored as the difference from the automatic framing, so it carries over to interfaces of other shapes and sizes.

The document opens with a **table of contents** of its own (the video's time-code contents has no meaning on paper): one line per chapter that actually prints — chapter number, name, a dotted leader and **the page the chapter starts on** — and, under each chapter, **a line per step**: its number, its name and the page it is on, one step in and a shade lighter. Switch the step lines off with **...and the steps under each chapter** in the document settings; the chapter lines stay. It is built by itself and stays right by itself: join or split pages, leave steps out, sync with a changed animation — the page numbers follow. A chapter whose steps are all left out of the document has no line. In the PDF every line is a link to its page. In a Hebrew document it reads *תוכן עניינים* and runs right-to-left.

- The contents is a real page and is counted: at the front it is page 1 and the first step page is page 2 (*Page 2 / 41*). It takes as many pages as its lines need (a chapter line is taller than a step line, and only the first page carries the title), and everything after it shifts accordingly.
- **Put it where it belongs.** A contents is rarely the very first page. Drag its  grip in the list — a blue line shows where it will land — or right-click it: **Move up / down**, **Move to the start / the end**. It is anchored *before a step*, so it keeps its place when you join, split or re-sync pages. Everything is renumbered, and its own lines follow.
- In the workspace it is a box of the list,  **Contents**: click it to see it in the centre; the arrows reach it like any other page. The list shows the **printed** page numbers.
- Switch it off with **Table of contents (chapters → pages)** in the left pane; the numbering then starts at 1. A project without chapters has no contents page.

For now the contents lists chapters only. Listing the steps of each chapter as well is planned, once it is clear how much detail a long manual can take on one page.

Custom pages

Not every page of a manual is made of steps: a cover, a safety notice, a parts list, a revision table. **Custom page** (above the list) adds an empty page right after the page you are on. It carries the

document's header, footer and watermark and is numbered like any other page; between the header and the footer it is yours.

- The bar over the page adds items: **+ Text** and **+ Picture ▾ — From a file...** (stored inside the project, at most 2400 px; its frame starts in the picture's own proportion) or **From the animation...**: a list of **every step**, chapter by chapter, each with its thumbnail. A step picture is not stored: it refers to the step and is rendered like the page pictures, so it follows the step when the animation changes. With it selected the bar offers **↗ Step N ▾** (another step) and **↳ Before** (the state the step starts from, seen from its camera, instead of its final state).
- **+ Table** adds a table — a parts list, a torque list, a specification.
 - **Type**: double-click a cell. **Tab** moves to the next cell, **Shift + Tab** back, **Esc** abandons that cell.
 - **Pick cells**: click one, or drag across several (**Shift** + click extends); click a **grip** to pick that whole row or column. Everything on the bar then acts on what you picked — add a row above or below it, copy the row, take a row or column away, or set the look of just those cells: **bold, italic, alignment, text size, text colour and a shade**.
 - **Right-click** a cell for the commands that belong to its row and column (add above / below / before / after, delete, merge); right-click a **grip** and the menu speaks only about that row, or only about that column.
 - **Merge**: pick a rectangle and press **■ Merge** — the texts join into one cell. **□ Unmerge** breaks it apart again.
 - **Move a row or a column**: drag the **blue grip** above a column or beside a row and drop it where the orange line shows. A row or column that runs through a merged cell cannot be moved — unmerge it first, and the app says so.
 - **Sizes**: drag the blue bar on a **column border** for its width, or on a **row border** for its height. A row you never touch is as tall as its text.
 - **Paste from a spreadsheet**: copy a block in Excel, **click the cell it should start at** (or double-click into it) and press **Ctrl + V** — tabs become columns, line breaks become rows, and the table grows to hold it. A single value lands in that one cell — and a **picture** on the clipboard lands *inside* the cell, scaled to fit it. If the copied block holds **pictures in its cells**, they come across with it: the spreadsheet also puts an HTML copy on the clipboard, and the pictures are read from it cell by cell. (A picture that *floats over* the sheet rather than sitting in a cell belongs to no cell, so it is not brought in — paste it into its cell on its own.)
 - **Copy cells**: pick a block and **Ctrl + C**. It goes to the clipboard as plain tab-separated text — paste it into Excel and it arrives as cells — and pasting it back into a table here brings each cell's **look and picture** with it, which plain text cannot carry.
 - **Several rows or columns at once**: pick them (drag across, or click one grip and Shift+click another) and the delete commands say so — *Delete rows 2–4*. **Col** copies a column the way **Row** copies a row, pictures and all.
 - The whole table still has its own **Header** row, **Grid** lines, **Stripes**, text size, alignment and colour. It keeps to its frame: make the frame taller (or the text smaller) if a row falls outside it.
- **Click** an item to select it, **drag** it to move it, drag a **handle** to resize it; arrows nudge by 1 mm (Shift = 5 mm); **Delete** removes it; **↑ Front** brings it above the others. Items snap to a half-millimetre grid and stay inside the content area.

- **Double-click a text** to type; click away to finish, Esc to abandon. Text wears the document's body font — the same one as the step texts, so the manual stays of a piece — and you choose the size (any number of points), bold, italic, alignment and colour. Hebrew and mixed lines are handled per box.
- A picture is cropped by its frame like the step pictures: **Fill**, **Whole**, **– / +**.
- Every gesture — a drag, a resize, a typed text, a delete — is one undo step.
- The page has a name for the list (left pane), a **⋮** grip to move it anywhere in the document, and a right-click menu: move, rename, delete. It is anchored before a step, exactly like the contents.

Tables, and a text box / table of this kind for the animation's overlay, are planned.

Your own page templates

When none of the prefab layouts fits, draw one. In the left pane, under the page's template, click  **New template....** The centre turns into a layout editor that starts from the current page's layout: the header and the footer are fixed grey bands, the **text box is blue**, the **picture frames are green** and numbered in the order the pictures fill them (the side panes are dimmed until you save or cancel).

- **Drag a box** to move it; drag a corner or an edge handle to resize it.
- **Drag on empty paper** to draw a new picture frame — or press **+ Picture frame**. Up to eight frames; a template with no frame at all (a text page) is allowed.
- The bar shows the selected box's **X, Y, W, H in millimetres** — type exact values — and, for a picture frame, ratio buttons (**16:9 3:2 4:3 1:1 3:4**: the width stays, the height follows), **◀ # / # ▶** to change which picture the frame receives, and **Delete frame**.
- Keyboard: arrows nudge by 1 mm (Shift = 5 mm), Tab goes to the next box, Delete removes the selected frame.
- Everything snaps to a half-millimetre grid and stays between the header and the footer, inside the side margins. **Boxes may not overlap**: overlapping boxes turn red, the bar names them, and **Save template** waits until they are apart. A frame does not need the video's proportions — pictures are cropped into it.

Save template asks for a name, stores the template **inside the project**, puts the current page on it and adds it to the template list of every page. A page that uses one of your templates offers  **Edit** (changes every page that uses it) and **Delete it** (those pages go back to the automatic layout). In a right-to-left document your frames are mirrored like the prefab ones, so draw them in left-to-right order.

Text

Every line starts as the step's voiceover and *follows* it: change the narration and the document changes with it. To rewrite a line for print, **click it on the page and type**; clicking away commits it, Esc abandons the change. From that moment the line is the document's own copy (*rewritten for the document*, blue ring on its number) — the voiceover is not touched, and the other way round. If the voiceover changes later, the line gets an amber ring and the left pane shows  **the voiceover changed after you rewrote this** with the new voiceover underneath, so you can decide: **keep mine** or **use the voiceover again**. Hebrew and other right-to-left lines are laid out right-to-left automatically, number badge included.

When a page has more text than its template has room for, the text area turns red with **& the text does not fit** — live, while you type. Shorten the text, split the page, or choose a template with a larger text area. Export asks before it writes a PDF with a cut page.

When the animation changes afterwards

Pages refer to steps by their permanent identity, not by position, so you can keep working on the animation. When the step list no longer matches the pages, the  **Sync** button turns amber and steps the document does not know yet appear dimmed in the list. Syncing never changes anything silently — every adjustment leaves an  on the page it touched, with a sentence saying what happened:

What changed in the animation	What sync does
A new step between two steps of the same page	It joins that page ().
A new step anywhere else	It gets a page of its own, in order ().
A step was deleted or hidden	It leaves its page (). A page left with no steps stays, marked $\emptyset$, until you delete it.
A step was moved to another place	It leaves its page () and is placed like a new step where it landed. A whole page that moved together simply moves, untouched.
A picture was pinned to a step that left the page	The picture is kept and flagged () — you decide.

Read the marks, fix what needs fixing, then  **Seen** — **clear these marks** (left pane) or  **Clear all** .

Hebrew and other right-to-left documents

Reading direction is always in the left pane, in the **Document** block under **Step numbers** — not only when the project is in Hebrew. It is *Automatic* by default: SBS counts the letters of everything the document prints, and when most of them are Hebrew or Arabic the whole document turns right-to-left — the header and footer mirror, lines and their number badges start from the right, picture numbers move to the top-right corner, and **the first picture of a page is the right-hand one** (the frames are mirrored across the page, so the reading order of the pictures matches the text). The built-in phrases print in the document's language (מהדורה ,12 מתוך 1 עמוד B, לפני) unless you re-worded them. A mixed line — Hebrew with a part number or a torque in Latin letters — is handled per line. Force **Left-to-right** or **Right-to-left** if the guess is wrong for your document.

Watermark

The left pane has a  **Watermark** section (click its title to open it). Tick **Print a watermark on every page** and choose the kind:

- **Text** — DRAFT, CONFIDENTIAL, a customer name; two lines are fine, Hebrew too. Size is in points and takes **any number**; pick a colour. The default is a mid grey because it reads both on the white

page and over a dark picture — black disappears on dark pictures, white on paper. In the PDF the mark is real, crisp text at any zoom.

- **Image** — a logo or a stamp. **Choose image...** opens a conversion window, because logos arrive in three shapes and a watermark needs real transparency: *artwork on a white background* (the white is made transparent), *artwork on a black background* (the black is made transparent), or *already transparent* (kept as it is). SBS guesses which one it is and shows the result over a white page and over a dark picture so you can judge it; **Make it one flat colour** turns a coloured logo into, say, a grey mark. The conversion is exact — soft edges stay soft, colours stay true — and is done once: the converted image is what the project stores and what every output uses. **Width** is a percentage of the page width.

Opacity and **Angle** are sliders that redraw the mark on the page while you drag (one undo step when you let go). **Where:** *Over everything — the pictures too* (the usual choice: the mark also protects the pictures) or *Behind the text and the pictures*. The watermark never gets in the way of editing: clicks go through it to the text and the pictures. It belongs to the document only — the animation and the video are not affected — and an image watermark is embedded once in the PDF however many pages there are.

Designing the header and the footer

Out of the box the header shows the logo (Header tab ▶  + **Logo**; a project that never defined one shows its first visible header image), the company, the title and *document no.* · *revision*; the footer shows the project name, the chapter and *Page n / N*. To design them yourself, **click the header or the footer on any page** — or  **Edit...** under *Header and footer* in the left pane. The page behind dims (it is only there to show the result; ← / → still walk the pages) and the band you are editing gets a dashed frame. The first time, the standard header and footer simply become movable items that look exactly the same — nothing changes until you change it. What you design is printed on **every page**: step pages, the contents, custom pages.

- **Header / Footer** on the bar choose the band — or click an item of the other one.
- Items work as on a custom page: click, drag, handles, arrows (1 mm, Shift 5 mm), **Delete**, **↑ Front**; text has size, bold, italic, alignment and colour and is centred vertically in its box. Items cannot leave their band (header 12–198 × 10–28 mm, footer 12–198 × 275–287 mm).
- **+ Field ▼** adds a value that fills itself in on every page: title, company, document no., revision, project name, chapter name, chapter number, page number, number of pages, date of the export. With a text selected the field is added *to that text*, so one box can read `Page {page} / {pages}`. On the page you see the values; **double-click** the text and you see what was written — the `{names}` — to edit it.
- **+ Picture ▼** adds **the project's logo** (it follows the project: change the logo there and it changes here; always shown whole) or a picture from a file (a certification mark, a second logo).
- **_ Line** shows or hides the rule of the band.
- On a right-to-left document the whole design is mirrored by itself — you design it once.
- **✓ Done** or Esc goes back to the page. **↶ Standard** (or *back to the standard one* in the left pane) throws the design away; undo brings it back.

The PDF

Title, company, document number and revision (left pane) feed the header's fields. ↓ **Export PDF...** writes the whole document (into the project's `exports` folder by default): real, selectable, searchable text — not a picture of text — with a bookmark per chapter. Every edit in the workspace is undoable with Ctrl+Z.

20. Settings, maintenance and housekeeping

Settings (Ctrl+,)

Section	Contents
Keybindings	Rebind the single-key actions (work camera, overlay edit, global mode, gizmo local/world, capture camera, overlay x-ray): click, then press the new key. Default restores the factory key.
Auto-backup / Autosave	Interval, how many backups to keep, Back up now .
Hardware insertion defaults	Nuts, spacing, gap scale, reposition pre-step (ms) used when a template does not override them.
Cloud TTS	Google Cloud Text-to-Speech API key; Test .
Preferred narration languages	Order of languages offered first in pickers.
Camera	Field of view and orbit defaults for new projects.
Default background	Solid colour for new projects' viewport.
CAD import (STEP / IGES)	Tessellation quality and default tree structure for imports.
Export defaults	Default output format, frame rate, step hold.
Translation engine	Server URL (OpenAI-compatible), model name, glossary, parallel requests.

Maintenance commands (Edit menu)

Recover stuck inputs

If a text field stops accepting keystrokes (rare, after certain popups), this releases it without a reload.

Rebuild Cascade

Recomputes the folder-to-part inheritance for every step from the stored data. Use after a big tree reorganisation if some step shows a stale hierarchy.

Clean up folders...

Finds wrapper folders with a single child across the loaded models, previews the flattening, and offers **Save first**, then **remove** or **Remove without saving**.

Model source transform...

Per-model scale (Metric / Imperial), rotation and offset applied at the source, with per-axis reset and **Reset all to identity**. Steps are not re-baked automatically — re-render afterwards.

Scan steps for what is in frame...

See 19.5.

Title Manager... / Unify Constant Titles / Clean Up Unused Constant Titles / Languages...

See 12.5 and chapter 13.

Reload versus restart

Ctrl + **R** reloads the app window and re-opens the current project from disk: the live scene is rebuilt from the saved data, which cures most "the picture looks wrong but the data is right" situations. Save first — the reload is not guarded. **Ctrl** + **Shift** + **R** also clears the window cache. A full restart of the application is only needed after installing a new version.

Memory

The app window has a hard memory ceiling of roughly 3.5 GB. Very large projects (hundreds of overlay images, long image sequences, many voices) approach it. Symptoms: slow undo, a white window, an export that dies mid-way. Remedies: keep overlay images small, store video as files (they are never inlined), **Clear inactive voices**, split the project into books and stitch the exports, and 🗑 **Clear history** in the ↶ tab on long sessions.

21. Troubleshooting

Symptom → what to do

Symptom	Likely cause	Do this
A part is missing or sits in the wrong place in one step, but only after jumping around	The live scene drifted from the data (an old bug, fixed in V0.3.3)	Save, then <code>Ctrl + R</code> . If it comes back, run <code>window.sbsDiag.visibilityAudit()</code> in the console and send the output.
Project opens with an asset warning	Model file moved or renamed	Browse... to relink, or Override to open with placeholders; fix the path later via <code>Ctrl + B</code> .
Narration is slow, silent, or the first play hangs	Voice engine still warming up, or GPU path failed and fell back to CPU	Wait for the warm-up message; then <code>await window.sbsTTSDiag()</code> in the console. <code>window.sbsTTS.forceCPU()</code> forces the CPU path for this session.
Text field ignores typing	Focus stuck after a popup typing	Edit ▶ Recover stuck inputs.
Steps 55–57 (say) always show ★ even after rendering	Something on those steps changes every time the project loads	<code>window.sbsDiag.stars(stepId)</code> lists the reason per step; send it.
Export re-renders everything after an app update	Fingerprint format changed	Choose Trust the stars once.
An imported video step is soft, blurry, pixelated, low resolution, low quality — worst on its final / last / held frame	The source segment was rendered at delivery quality (Medium 4 Mbps) and is compressed again here; a short hold after motion leaves the last frame unrefined	In the source project: Export quality High or Very High , re-render that step, import again. See the video-step note.
Instant step on a chapter start plays a half transition	Old instant block format	Re-pick the easing from the dropdown on that step.
Duplicate flat shapes vanish after reload	Known, rare	<code>window.sbsDiag.shapes()</code> then <code>window.sbsFix.pruneShapes()</code> ; report the console output.
White window, or export dies mid-way	Memory ceiling	See 20.4.

Console diagnostics

Open Developer Tools with `Ctrl + Shift + I` and pick **Console**. Everything under `window.sbsDiag` only reads and reports; everything under `window.sbsFix` repairs and is undoable or asks first. Paste the output when reporting a problem.

Command	Reports
<code>window.sbsDiag.stars() / stars('step_id')</code>	Which steps are starred and why (reference, order, definition, animation, settings, node, tool).
<code>window.sbsDiag.frameVis('step_id') / await window.sbsDiag.scanFrames()</code>	The in-frame record of a step; run the scan from the console.
<code>window.sbsDiag.visibilityAudit() → visibilityRepair()</code>	Parts whose effective visibility disagrees with the step data.
<code>window.sbsDiag.verifyHome()</code>	Objects whose live home position disagrees with the stored data.
<code>window.sbsDiag.prims() → window.sbsFix.prims()</code>	Primitive parameter health.
<code>window.sbsDiag.shapes(), cablesAudit(), cableSteps(), iface(), videos(), archived(), rmHealth()</code>	Per-feature audits: flat shapes, cables, interfaces, video overlays, archived nodes, replace-models.
<code>window.sbsDiag.fadeTrace(), animTrace(), cableTrace(), traceSelected(), traceOff()</code>	Live traces of a transition for the selected objects; turn off with <code>traceOff</code> .
<code>window.sbsDiag.input() → window.sbsFix.input()</code>	Stuck input diagnosis and release.
<code>await window.sbsTTSDiag()</code>	Voice engine: device, warm-up, cache, last error.
<code>window.sbsMemReport(), window.sbsFreeMemory()</code>	Heap usage by category; drop caches.
<code>window.sbsRenderPlan(), window.sbsCachePurge()</code>	What the next export would render; purge stale clips.

22. Keyboard shortcuts

Shortcuts follow the physical key, not the letter it types. They work the same whichever keyboard language is active — leave the keyboard on Hebrew after typing a title and **F**, **Ctrl + S**, **Ctrl + C** / **V** / **D** and **Ctrl + Z** still do their job. The single-letter ones (**F**, **W**, **O**, **L**, **C**, **X**, **M**, **Space**) can be moved to another key in **Settings ▶ Keybindings**.

Project

Ctrl + N	New project
Ctrl + O	Open
Ctrl + S / Ctrl + Shift + S	Save / Save As
Ctrl + B	Browse assets
Ctrl + ,	Settings
Ctrl + Z / Ctrl + Y	Undo / Redo
Ctrl + X C V A	Cut, copy, paste, select all (context-aware: tree, steps, overlay)
Ctrl + Alt + U	Unify constant titles
Ctrl + R / Ctrl + Shift + R	Reload / hard reload the window
Ctrl + Shift + I	Developer tools (console)

Viewport (rebindable in Settings ▶

Keybindings)

W	Work camera on / off
C	Capture the view into the step camera
Alt + wheel	Perspective: wide-angle ↔ orthographic (the framing is held)
Middle-drag	Pan — what you grab stays under the cursor at every perspective
F	Fit selection (or all)
L	Gizmo local / world
Space	Global mode on / off
O	Overlay edit on / off
X	Overlay x-ray
M	Overlay magnet (snapping) on / off
Shift while dragging an overlay item	Move straight along X or Y
Alt while dragging an overlay item	No magnet for that moment
Esc	Cancel tool / deselect / close popup
Delete / Backspace	Delete the selected step, overlay item or node (with confirmation where destructive)

Ctrl + D	Duplicate whatever is selected and has a duplicate: overlay items (text boxes, shapes, pictures) while editing the overlay; otherwise the 3D selection — screws (same template), primitives, flat shapes — each set beside its original; with nothing selected in the scene, the selected steps (or the active step), each copy right after its original. One undo entry each
Ctrl + Shift + D	Duplicate in place — the same, but the copy sits exactly on top of the original (also in the overlay's right-click menu)
← / →	Previous / next step
Shift / Ctrl / Alt + click	Add / toggle / invert selection

23. Glossary

AL1 / AL2

Animation-length variables (ms) that presets reference; AL1 is the camera's home, AL2 the objects'.

Archive

Remove a part from every step without deleting it.

Cascade

The inheritance of visibility and transforms from folders to their contents inside a step.

Constant (title / position)

An overlay item defined once and shown at the same place on many steps.

Fingerprint

The hash of everything that influences a step's rendered frames; the render cache key.

Frame record

The list of parts that are actually inside a step's picture, produced by the frame scan.

Global mode

Transform edits carry into every step.

Instant block

A zero-length phase pinned to the top of a step's animation that holds every channel: a cut.

Language pack

The `.sbslang.json` sidecar holding one language's texts and voice choice.

Pivot

The point a part rotates around; also the centre of its animated arc.

Preset (colour)

A named material (colour, opacity, solidness, metalness, roughness) worn by parts.

Replace-model

A node that can show one of several alternative geometries per step.

Segment / clip

The cached render of one step (its transition and hold).

Snapshot

The complete scene state stored on a step.

Solidness

X-ray strength of a preset; below 1 the part becomes see-through.

Star ★

"Changed since last rendered" marker on a step; chapters sum their stars.

Template (camera / hardware / note)

A named definition that instances follow; edit once, apply everywhere.

Work camera

A scratch camera (W) that does not touch the step's saved view.